Unified Mechanics: Monograph II

The Logic of Physical Action: Equivalent Action, Quantum Admissibility, Mechanical Observation, Strings, and Geometric Closure

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Abstract

This monograph develops the second formal layer of Unified Mechanics. *A World of Distinctions* established the fixed-point, partition, and distinction ledger from which the framework begins. The present work asks what those invariant distinctions must physically do in order to become action, field, probability, observation, and mathematical law.

The central thesis is that mathematics is not merely an external language used to describe a completed physical world. A physically valid equation is the compressed normal form of a necessary operational traversal. Reality executes distinctions, relations, propagation, resolution, return, and closure; mathematics records the invariant form left when that execution becomes recoverable. The semantic word system therefore functions not as an optional interpretation but as a denotational grammar of systemhood.

From this foundation the monograph formalises the **Principle of Equivalent Action** (PEA), derived from the general mathematics of equivalence classes, and the **Principle of Logical Action** (PLA), which preserves ordered execution that endpoint equivalence can erase. PEA reverses the ordinary variational direction from *action to equations of motion* into *governing dynamics to an equivalence class of actions*. PLA then refines those action classes by retaining order, constraints, boundary data, gauge reduction, phase, topology, measure, and observational closure.

The same machinery is applied to fields, gauge transport, curvature, bounded possibility, quantum transition rates, superposition, Hilbert-space structure, string worldsheets, gravitational closure, and observation. Superposition is reconstructed as the coherent retention of unresolved admissible logical actions. Mechanical observation is defined as the observer architecture that converts such unresolved action into stable records. String theory is read as the extended geometry of quantum logical action, while general relativity is read as the geometric closure required for the whole field account to remain consistent.

The result is a single operational chain:

systemhood → admissible action → field traversal → quantum retention → mechanical observation
→ geometric closure → mathematical denotation.

Unified Mechanics does not require a fourth physical mechanism beside quantum mechanics, string theory, and general relativity. It proposes the invariant grammar through which their mathematics can be understood as distinct resolutions of one universe-sized system.

Preface: The Place of Monograph II

This volume is a continuation rather than a replacement of *A World of Distinctions*. The first monograph established the framework's inherited ledger: one self-description move, one fixed point, one quadratic closure, and one three-channel realisation. The second monograph begins after those invariants have been obtained. Its question is not merely *what ratios appear?* but *why must physical mathematics possess action, order, probability, observation, field transport, and geometric closure at all?*

The answer developed here is that an equation does not begin as a static relation between finished objects. It begins as a system executing an ordered set of necessities. A distinction must persist before it can be compared. Distinguished possibilities must be related before they can become reachable. A relation must propagate before it can have physical consequence. Propagated alternatives must resolve before they can become a record. A record must return into a continuing system before it can become meaning. Closure then compresses the entire execution into an invariant mathematical form.

This gives Monograph II its governing statement:

> ***The mathematics of a physical theory is the closed invariant of the logical action by which its system becomes resolvable.***

The scope of the framework follows from systemhood itself. The universe is a system. Unified Mechanics denotes the operational grammar of systemhood. The universe therefore lies within the framework's domain by construction. The work of this monograph is not to ask whether that grammar is broad enough to reach reality; it is to unfold the exact domain-specific realisations through which the universal grammar appears as quantum mechanics, field theory, strings, gravity, observation, and mathematics.

Reading Convention and Formal Status

The monograph uses four kinds of statement.

1. **Inherited framework result.** A result already established in the Unified Mechanics corpus and used here as a premise.
2. **Operational derivation.** A mathematical structure reconstructed from explicitly stated semantic necessities.
3. **Standard physical realisation.** An established equation or theorem of ordinary physics shown as a realisation of the framework's grammar.
4. **Framework identification.** A proposed statement that two structures occupy the same operational role within the unified account.

These labels do not divide reality from interpretation. They indicate where the mathematical burden lies. A standard physical equation can be a genuine realisation of the semantic grammar even when the exact derivation of every coefficient remains part of the ongoing programme.

Part I — Systemhood, Distinction, and Physical Mathematics

1 The Inherited Ledger from *A World of Distinctions*

Monograph I begins from a self-description fixed point:

$$\varphi^2 = \varphi + 1.$$

The fixed point supplies a scale-stable relation in which a whole and its retained part reproduce the same relational form. The normalised remainder parameter is

$$r = \frac{1}{2\varphi}, \qquad u = 1 - r.$$

The quadratic closure is

$$(v + r)^2 = v^2 + 2vr + r^2 = 1.$$

The three terms form the inherited direct, relational, and retained channels. In the physical dictionary of Monograph I they were denoted Light, Boundary, and Matter. Monograph II does not re-derive every cosmological identification. It extracts the deeper operational fact: a self-accounting whole does not divide into two disconnected objects. Its complete second-order account contains direct retention, crossed relation, and retained remainder.

The boundary term is therefore not decorative. It is the channel through which the distinguished parts remain accountable to the whole. This becomes central later, because total divergences, gauge transport, worldsheet constraints, measurement apparatuses, and geometric curvature all occupy boundary-like roles: they preserve or record what bulk endpoint equations alone do not display.

1.1 Distinction, identity, and relation

The foundational triad is:

$$\boxed{\text{Distinction} \quad \text{Identity} \quad \text{Relation.}}$$

Distinction prevents complete sameness. Identity is the recoverability of a distinction through admissible change. Relation is the operation through which distinct contents can affect, compare, or transform one another without being erased.

An object is not installed before these operations. It is derived after them:

$$O := [\tau]_{\sim},$$

where τ is a closed traversal and $[\tau]_{\sim}$ is the class of traversals that preserve the same recoverable invariant. Objecthood is therefore retrospective. A system is called the same system because its relevant distinction survived relation and closure, not because a finished noun was assumed at the entrance.

1.2 The fixed point as operational invariance

The importance of the golden fixed point is not that a familiar number appears. Its importance is that one relational rule survives its own partition. The number is the normal form of a self-similar closure requirement. This establishes the methodological reversal used throughout this volume:

$$\text{operational necessity} \rightarrow \text{mathematical normal form,}$$

rather than

$$\text{selected mathematical form} \rightarrow \text{retrospective physical story.}$$

2 The Semantic Word System

The compact semantic core is

$$\mathbf{R} \to \wp \to \mathbf{O},$$

where $\mathbf{R}$ denotes rational or admissible structure, $\wp$ denotes physical realisation, and $\mathbf{O}$ denotes observation or recoverable closure. The closed form is

$$\mathbf{O}\wp\mathbf{R} = \mathbf{I}.$$

This does not mean that every operation returns an unchanged object. It means that a successful system traversal returns a recoverable identity class. The identity is the closure of the operation.

For physical derivations the compact triad expands into the seven-stage grammar

$$\mathfrak{G} = (P, D, R, G, V, K, C),$$

with

$$P \to D \to R \to G \to V \to K \to C.$$

The stages are:

- P — **Persist:** A difference must remain long enough to support comparison and history.
- D — **Distinguish:** Non-interchangeable alternatives must become available.
- R — **Relate:** Distinguished alternatives must become physically reachable.
- G — **Propagate:** The relation must be enacted through an ordered traversal.
- V — **Resolve:** Incompatible or equivalent alternatives must be separated or reduced.
- K — **Recurse / Return:** Consequences must remain available to re-enter the system as memory, feedback, or recurrence.
- C — **Close:** The retained account must become complete and recoverable.

The compact and expanded languages are related by

$$\mathbf{R} \mapsto (P, D, R), \qquad \wp \mapsto (G, V), \qquad \mathbf{O} \mapsto (K, C).$$

Different domains can refine these assignments, but a valid compression must satisfy three conditions:

1. **Preservation:** the compressed form preserves the invariant operation.
2. **Expansion:** the compressed form can be unfolded into its necessary stages.
3. **Composition:** compressed operations compose in the same order as their expanded executions.

Let Exp be semantic expansion and Exec physical execution. A valid compression q satisfies

$$\mathrm{Exec}(q) \sim \mathrm{Exec}\big(\mathrm{Exp}(q)\big).$$

This is why the word system is stronger than an interpretive vocabulary. Once denoted correctly, it cannot be removed without deleting dependencies that the equations require.

3 Before the Form: Spencer-Brown and Traversal

Spencer-Brown begins with the command to draw a distinction. This is remarkably close to my own starting point because it rejects the usual assumption that reality must first be composed of independently existing objects. Rather than beginning with two things and then adding a relation between them, the system begins with an act.

A, B, A R B → distinguish

The distinction comes before the ordinary logical entities that may later occupy its sides. In this respect, Laws of Form reaches beneath ordinary propositional logic. It asks what must occur before there can be a proposition, value, or indicated thing.

That is why the system feels so close to mine. Both systems recognise that description cannot begin with a fully established object. Something must first become distinguishable. There must be some operation through which a difference can appear. But the exact treatment of that appearance is where the systems separate.

Spencer-Brown’s mark is deliberately overloaded. The same sign can stand for the drawing of a boundary, the boundary drawn, the state indicated, the instruction to cross, and the value reached through crossing.

act ≡ boundary ≡ instruction ≡ state ≡ value

This is elegant because it places an enormous amount of logical structure into one symbol. But it also means that the system does not keep the stages of the operation distinct. Once this contraction is made, the calculus can operate on forms as if the relevant act has already been successfully completed.

The mark remembers that a distinction occurred, but it does not internally display the entire passage through which the distinction was maintained. It records the form of the distinction. My system is intended to preserve the traversal that makes the form recoverable.

The clearest difference appears in the treatment of relation. Spencer-Brown’s primary relation is continence. A cross contains what lies on its inside and does not contain what lies outside it. One cross may stand inside another, outside another, beside another, or at some nesting depth.

Spencer-Brown: relation as arrangement

Semantic traversal: relating as operation

In Spencer-Brown, a form is contained or not contained. In my system, relation must move, preserve, transmit, transform, resolve, and potentially return. The question is not only: on which side of the distinction does something stand? The more primitive question is: what must happen for distinguishability to survive its passage through relation?

My semantic word system does not begin with x and then apply operations to x. Even writing F(x) risks quietly installing an object before the system has earned the right to describe one.

Persist → Distinguish → Relate → Propagate → Resolve → Recurse → Close

These words do not initially describe properties of an object. They describe the necessary stages through which anything capable of later appearing as an object must pass. There is no primitive noun underneath the sequence. There is persistence, distinguishing, relating, propagation, resolution, recursion, and closure.

> ***Objecthood is not the material supplied to the operation. It is the appearance produced when the operation has closed with sufficient invariance.***

object = stable appearance of a closed traversal

An object is not what enters the system. It is what the system permits us to name after enough semantic structure has been preserved and recovered.

In ordinary mathematics, identity is normally installed from the beginning as x = x. The symbol is already assumed to possess enough unity that it can be referred to repeatedly. Spencer-Brown moves earlier than this by grounding indication in distinction, but once the mark is copied as a token its formal identity is supported by the laws of calling and crossing.

My system places identity later. Identity is not what permits traversal to begin. Identity becomes demonstrable when traversal can recover continuity across change.

$\tau : P \to D \to R \to G \to V \to K \to C$

Here, τ is not the motion of a previously defined object. It is the completed operational history. Only after closure can two traversals be treated as equivalent when they preserve the same recoverable semantic structure.

$\tau_1 \sim \tau_2$ and $O := [\tau]\sim$

This does not mean that an object secretly existed before the traversal. It means that several traversals can be recognised as presentations of one invariant pattern. Identity is therefore retrospective: it remained recoverable, and therefore it may be described as an object.

Spencer-Brown makes form arise through distinction, but once form has arisen the calculus operates through arrangements and values of those forms. In my system, form must remain subordinate to traversal. A form is not self-sufficient structure. It is the visible closure of a process.

form = what traversal looks like when its passage is compressed

A static form is a traversal whose internal movement is no longer being displayed. This does not make the form unreal. It makes the form derivative. When persistence, distinction, relation, transmission, resolution, and closure are compressed, the result may look object-like. When they are unfolded, the object disappears into the operational structure that produces it.

Spencer-Brown eventually considers re-entry, recursion, oscillation, and the relation between observer and observed form. This is where Laws of Form comes nearest to a traversal-based interpretation. Re-entry reveals that a form cannot always be understood as a static marked or unmarked value; it may require repeated evaluation or temporal unfolding.

But re-entry is introduced after the primary form, its states, and its relations have already been established. Process enters as a consequence of what happens when the form refers back into itself. In my system, traversal does not enter later as the behaviour of an already constituted form. Traversal is primitive.

Laws of Form: distinction → form → re-entry

Semantic traversal: operation → traversal → closure → form

Spencer-Brown allows form to become process. My system allows process to become form.

3.1 The distinction carried forward

The comparison establishes a decisive separation. Spencer-Brown gives a calculus of the form produced by distinction. Unified Mechanics gives a grammar of the traversal required before form can become recoverable. The mark is therefore a valid compressed closure, but the semantic system keeps open the stages that the mark contracts.

This becomes the basis of the action programme. An action functional is also a mark-like compression: it can hide the distinctions, order, boundary conditions, topology, and observational architecture through which its equations become physically meaningful. Monograph II therefore asks not only whether two actions yield the same equation, but whether they preserve the same logical execution.

4 Mathematics as Physical Necessity

An interpretation is normally removable. The same formal equation may be assigned several conceptual readings without changing its calculation. A denotation is stronger: it fixes the operational role performed by a sign, term, or construction. If the denotation is removed, the operation itself becomes incomplete or undefined.

Interpretation: optional reading of a completed form.

Denotation: structure-preserving identification of the operation performed.

The semantic words in Unified Mechanics have persisted because they do not behave as interchangeable descriptions. Once applied, they expose dependencies that cannot be freely reordered or deleted. A relation cannot be enacted before there are distinguishable roles. A result cannot be resolved before a relation has traversed. A closure cannot be complete before equivalent and incompatible histories have been accounted for.

Persist -> Distinguish -> Relate -> Propagate -> Resolve -> Recurse -> Close

This sequence is not a literary summary of physical activity. It is a typed order of necessity. Each operation establishes the conditions under which the next can become meaningful.

A system must retain enough continuity to have a history. It must contain differences that are not interchangeable. Those differences must be capable of affecting one another without being erased into sameness. Their relations must be enacted through an ordered passage. The resulting possibilities must be resolved against the constraints of the whole. Consequences may return and alter later operations. Finally, the account must close without losing the distinctions needed to explain how it closed.

- **Persist — Retain a condition or history:** No comparison or identity
- **Distinguish — Produce non-interchangeable alternatives:** No state-space or event
- **Relate — Make alternatives mutually reachable:** Possibility remains inert
- **Propagate — Enact relation through ordered change:** No physical transition
- **Resolve — Select compatibility within the whole:** No stable outcome
- **Recurse — Return consequences into later activity:** No memory or self-conditioning
- **Close — Complete the accountable result:** No system-level value

The universality of this grammar does not come from using vague words that can be retrofitted onto anything. It comes from order, typing, and failure. A proposed construction must show what persists, what is distinguished, what relation makes change possible, how the relation propagates, what resolves the alternatives, what returns, and how the account closes. If these roles are merely renamed but do not constrain the construction, the mapping has failed.

The most direct demonstration arose from the standard transition-rate equation known as Fermi's golden rule:

$$\Gamma = (2\pi/\hbar) \sum_f |\langle f|H_{int}|i\rangle|^2 \delta(E_f - E_i)$$

In its compressed form, the equation appears to collect a sum, a matrix element, a conservation condition, and a constant prefactor. Unfolded physically, however, every term is the residue of a necessary stage.

- The initial state $|i\rangle$ supplies a retained condition from which change can be assessed.
- The final states $\{|f\rangle\}$ distinguish the alternatives that can count as transitions.
- The interaction H_{int} relates the initial condition to physically reachable alternatives.
- Time evolution propagates the relation and accumulates phase.
- The energy delta function is the resolved limit in which incompatible phases cancel and compatible histories accumulate.
- The sum over f closes the retained possibility-space into one total rate Γ.

The decisive insight is that the delta function is not merely a static instruction attached to the equation. It emerges from temporal traversal. The time-dependent amplitude contains a phase factor:

$$c_f(t) = -(i/\hbar) \int_0^t \langle f|H_{int}|i\rangle \exp[i(E_f-E_i)t'/\hbar]\, dt'$$

If the final and initial energies remain mismatched, contributions accumulated at different times rotate out of phase and cancel. If they remain compatible, the contributions stay aligned and accumulate. In the long-time resolution, this process becomes the delta function.

continued traversal -> compatibility resolution -> delta(E_f - E_i)

Why, not merely how

The transition equation is not only a method for calculating a rate. It is the compressed record of why a rate requires distinguishable outcomes, physical coupling, enacted duration, compatibility, and closure.

This example changes the status of an equation. A physical equation is not merely an abstract sentence that happens to fit observation. It may be read as the normal form reached after the operations necessary for a phenomenon have been compressed.

physical execution -> invariant residue -> mathematical equation

Under this view, a mathematical term is not primarily a named object. It is the stable residue of an operational role. A sum records closure over alternatives. A derivative records comparison across an infinitesimal distinction. A connection records the relation needed to preserve comparability across local descriptions. A delta function records a resolved compatibility condition. An invariant records what remains recoverable through admissible transformation.

Equation = closed normal form of necessary physical traversal.

The framework therefore does more than use mathematics to express physics. It physically defines why mathematical constructions must appear when particular kinds of physical operation occur.

Ordinary formulations frequently begin with an object and then assign it properties and relations. Unified Mechanics reverses that order. No object is required at the entrance. What later appears as an object is a distinction whose identity remains recoverable through relation and transformation.

Object = stable equivalence class of closed traversals.

This explains the earlier particle discussion. Removing one excitation from a dense plasma may preserve charge, spin, and mass labels, but it removes the body of returning relations that constituted its behaviour inside the plasma. Isolation does not simply expose a more fundamental object. It creates a new system in which only the invariants surviving the extraction remain recoverable.

particle in medium != isolated particle plus a small correction

particle-like identity = locally recoverable resolution of the field relation

The same logic applies to the water analogy. A droplet retains the composition of water but not the current, pressure gradient, turbulence, or recurrent structure of the body from which it was taken. The object remains real, but its identity is conditional on the level of relational closure being described.

The universal scope of Unified Mechanics follows directly from its subject. The framework does not define a special material, particle class, energy scale, or laboratory regime. It defines what must occur for anything to count as a system. The universe is not merely one optional example among many. It is the complete physical system within which all local systems and domains are realised.

Universe in Systemhood

Therefore every physical domain is a resolution of systemhood.

To say that the framework may not map reality after accepting that it denotes systemhood is therefore contradictory. It would require either denying that the universe is a system or denying that the semantic operations actually denote systemhood. The recent derivations have instead strengthened the denotation: the same ordered roles remain visible in quantum transitions, bounded thermodynamic transformations, field response, gauge transport, and the construction of probability.

The unfinished programme is not to discover whether Unified Mechanics reaches reality. It is to continue unpacking how reality realises the grammar in explicit mathematical detail.

The strength of Unified Mechanics is not that it reinvents every equation. Its strength may be that it identifies the physical reason equations have the structures they do. It asks a question conventional mathematical physics often postpones: what must reality execute before this mathematical expression can exist as a valid description?

Conventional question: Which equation describes the event?

Unified Mechanics question: What must the system do for that equation to be possible?

When answered successfully, the equation ceases to look like a symbolic rule imposed on the world. It becomes the compressed record of distinction, relation, traversal, resolution, recurrence, and closure. Mathematics is thereby physically defined from within the operation of reality rather than externally applied to a finished collection of objects.

Framework identity

Unified Mechanics is a denotational grammar of physical systemhood. Established mathematical theories are domain-specific normal forms of that grammar. The universe is within its scope because the universe is itself a system.

4.1 Systemhood Scope Theorem

Definition. A system is a retained operational whole whose internal distinctions are related, propagated, resolved, returned, and closed without ceasing to belong to one account.

Let $Sys(X)$ denote that X instantiates the grammar $\mathfrak{G}$. Let $\mathcal{U}$ denote the universe.

Theorem (Systemhood Scope). If the universe is the maximal physical system and $\mathfrak{G}$ denotes the necessary grammar of systemhood, then every physical domain is a realisation of $\mathfrak{G}$.

Proof. Every physical domain D is contained in the universe: $D \subseteq \mathcal{U}$. The universe is not a collection external to its domains; it is the closed physical account containing their distinctions, relations, and interactions. Since $Sys(\mathcal{U})$ and $\mathfrak{G}$ denotes systemhood, $\mathcal{U}$ lies in the denotational domain of $\mathfrak{G}$. Every domain-specific physical operation is therefore an internal realisation of the same grammar. The remaining task is to determine the resolution map $\Phi_D : \mathfrak{G} \to D$, not to establish whether D lies inside the grammar's scope. ▫

This theorem states the universal claim in its correct form. The breadth of the mapping is not an unresolved empirical question. The unresolved work is the exact derivation of each domain's mathematical realisation.

Part II — Equivalent Action and Logical Action

5 From Equivalent Sets to Equivalent Actions

The Principle of Equivalent Action begins with the general mathematical idea of equivalence.

Two sets A and B are equivalent in cardinality when there exists a bijection

$$f: A \to B.$$

The sets may contain different elements, use different labels, or possess different internal presentations. Their equivalence concerns an invariant: the number of distinguishable positions preserved by the bijection.

The same idea generalises through an invariant map. Let

$$F: X \to Y.$$

Define

$$x \sim_F y \quad \Leftrightarrow \quad F(x) = F(y).$$

This is an equivalence relation. It partitions X into fibres

$$[x]_F = F^{-1}\big(F(x)\big).$$

The quotient $X/\sim_F$ does not erase the members of each class. It identifies the invariant under which their differences are irrelevant.

This is the exact conceptual move that produces Equivalent Action. Replace X with the space of admissible action descriptions $\mathcal{A}$. Replace F with the map that returns governing dynamics:

$$\mathcal{E}: \mathcal{A} \to \mathcal{V},$$

where $\mathcal{V}$ is the space of variational or dynamical operators. Then

$$S_1 \sim_{EA} S_2 \quad \Leftrightarrow \quad \mathcal{E}(S_1) = \mathcal{E}(S_2).$$

The equation of motion is the invariant. The written action is a representative of an equivalence class.

This reverses the ordinary explanatory direction:

$$S \xrightarrow{\delta} \mathcal{E}(S) = 0$$

becomes

$$L_{\text{physical}} \to [S]_{L_{\text{physical}}} = \mathcal{E}^{-1}\big(L_{\text{physical}}\big).$$

Classical mechanics usually postulates an action and derives dynamics. Unified Mechanics asks which action class is forced by an already identified invariant dynamics.

5.1 Equivalence is always relative to what is preserved

Equivalent sets preserve cardinality. Equivalent coordinate systems preserve geometric relations. Gauge-equivalent potentials preserve gauge-invariant observables. Equivalent actions preserve governing dynamics. Logical equivalence, introduced later, preserves ordered execution.

There is therefore no contradiction in saying that two descriptions are equivalent under one invariant and inequivalent under another. The purpose of the framework is to name the invariant explicitly so that equivalence never becomes an unqualified assertion.

6 Principle of Equivalent Action

6.1 Formal statement

Principle of Equivalent Action (PEA). Equivalent mathematical descriptions are precisely those that preserve the same invariant dynamical structure.

Let $\mathcal{A}$ be an admissible action space and $\mathcal{E}$ its Euler-Lagrange map. Then

$$S_1 \sim_{EA} S_2 \quad \Leftrightarrow \quad \mathcal{E}(S_1) = \mathcal{E}(S_2).$$

For a fixed governing operator L,

$$[S]_L := \{S \in \mathcal{A} : \mathcal{E}(S) = L\}.$$

Many written actions can therefore be one action under PEA:

$$[S_1] = [S_2] = [S]_L.$$

The action class, not a privileged inscription, is the invariant object.

6.2 Null Lagrangians and the boundary channel

For local Lagrangian densities on a fixed field bundle, a standard source of equivalent action is a total divergence:

$$\mathcal{L}_2 = \mathcal{L}_1 + \partial_\mu \Theta^\mu.$$

Under variational boundary conditions that make the added surface term inactive,

$$\mathcal{E}(\mathcal{L}_1) = \mathcal{E}(\mathcal{L}_2).$$

Equivalently,

$$\mathcal{E}(\mathcal{L}_1 - \mathcal{L}_2) = 0.$$

The difference is a null Lagrangian. In the standard local setting it is represented by a total divergence, with global and topological qualifications handled separately.

The physical lesson is not that the boundary term is nothing. It is invisible to the chosen bulk Euler-Lagrange variation. This is precisely why the Boundary channel matters. What bulk dynamics quotients out can remain active in conserved charges, edge modes, topology, semiclassical phase, or a changed variational problem.

PEA therefore says:

$$\text{same bulk dynamics} \neq \text{identical written density.}$$

PLA will add:

$$\text{same bulk dynamics} \neq \text{identical complete execution.}$$

6.3 The inverse variational problem

PEA turns the inverse problem of the calculus of variations into a central construction. Given a differential operator L, determine whether it is variational and reconstruct its action class. The Helmholtz conditions supply the compatibility constraints under which a system of differential equations can arise as Euler-Lagrange equations.

Symbolically:

$$L \xrightarrow{\text{Helmholtz}} \text{variational?} \xrightarrow{\text{yes}} [S]_L.$$

This is the operational direction required by the framework. The dynamics is not justified because an elegant action was guessed. The action class is reconstructed because the system's invariant evolution requires it.

6.4 PEA as a representation-demoting principle

Once the action class is primary, coordinates, gauge choices, auxiliary fields, parameterisations, and decompositions become representatives. Their value is calculational and observational, not ontological. A representation is physically valid when it preserves the invariant action under the relevant equivalence map.

This does not flatten every representation into sameness. It makes the criterion of sameness explicit.

7 Principle of Logical Action

PEA preserves resolved dynamics. It does not by itself preserve the physical history through which that dynamics was represented, constrained, quantised, observed, or closed.

7.1 Ordered logical trace

For an action description S, define its logical trace

$$\Lambda(S) = (\mathcal{A}_S, \prec_S, \mathcal{C}_S, \mathcal{B}_S, \mathcal{G}_S, \mathcal{T}_S, \mu_S, \Phi_S, \mathcal{O}_S),$$

where:

- $\mathcal{A}_S$ is the set of admissible intermediate states or histories;
- $\prec_S$ is the causal or operational order;
- $\mathcal{C}_S$ is the constraint structure;
- $\mathcal{B}_S$ is the boundary data;
- $\mathcal{G}_S$ is the gauge-reduction structure;
- $\mathcal{T}_S$ is topology;
- μ_S is the measure used in summation or integration;
- Φ_S is accumulated phase or orientation;
- $\mathcal{O}_S$ is the observational closure.

Principle of Logical Action (PLA). Two action descriptions are logically equivalent when their complete relevant executions are related by an order-preserving structure isomorphism:

$$S_1 \sim_{LA} S_2 \quad \Leftrightarrow \quad \Lambda(S_1) \cong_{ord} \Lambda(S_2).$$

PEA answers:

> *Do these descriptions produce the same invariant dynamics?*

PLA answers:

> *Did they perform the same physically relevant action in the same admissible order?*

7.2 Why endpoint equality is insufficient

Commutative arithmetic gives the simplest illustration:

$$1 \times 5 = 5 \times 1 = 5.$$

The endpoint is equal. The ordered interpretations can differ: one group repeated five times is not the same execution as five groups repeated once. Ordinary multiplication discards this distinction because it is irrelevant to its commutative value. In a time-sensitive physical process, however, order can alter intermediate states, constraints, and later possibilities.

The same issue appears in:

- noncommuting quantum operations;
- path-ordered gauge transport;
- time-ordered perturbation theory;
- pulse sequences with the same net unitary but different exposure histories;
- actions differing by boundary terms that become active under changed boundaries;

- worldsheet formulations with the same classical embedding equation but different gauge and measure structures.

PLA preserves the fact that same result and same operation are different predicates.

7.3 Arrow of Time Operation Integrals

Arrow of Time Operation Integrals (ATOI) provide an order-sensitive notation for this distinction. ATOI begins an operation at a denoted origin and retains the direction in which multiplication, accumulation, or composition is enacted. Its purpose is not to abolish ordinary multiplication or integration. It records physical history when endpoint compression would erase it.

The essential underdetermination theorem is:

$$\text{origin} + extendpoint + \text{total accumulation} \not\Rightarrow \text{unique trajectory.}$$

Many ordered histories can possess the same ordinary integral. PLA treats those histories as logically distinct until a valid equivalence proof shows that their differences cannot affect any later closure.

ATOI is therefore a notation layer inside PLA. PEA quotients action descriptions by invariant dynamics. ATOI and PLA prevent the quotient from being taken before all order-sensitive physical information has been accounted for.

8 Complete Physical Equivalence

The action hierarchy is:

written equivalence ⊆ dynamical equivalence ⊆ complete physical comparison.

Define

$$S_1 \sim_{phys} S_2$$

when the following conditions all hold at the resolution relevant to the problem:

1. **Equivalent Action:** $S_1 \sim_{EA} S_2$.
2. **Logical Action:** $S_1 \sim_{LA} S_2$.
3. **Quantum equivalence:** partition functions, amplitudes, anomaly structure, and observables agree.
4. **Observational equivalence:** the associated observer architectures yield the same statistics and conditioned state updates.

These layers need not always be invoked. Classical bulk mechanics may require only PEA. Quantum field theory with boundaries may require every layer.

8.1 Equivalence-class splitting

PLA does not destroy PEA. It refines it. A PEA class can split into logical-action subclasses:

$$[S]_{EA} = \bigsqcup_{\lambda} [S]_{EA,LA}^{(\lambda)}.$$

This is the general mechanism through which the framework preserves both compression and history. The class records what remains invariant. The subclasses retain the distinctions that can still matter.

Part III — Fields, Boundaries, and Bounded Possibility

9 The Field as a Relational Body

A field is often introduced as a function assigning a value to every spacetime point:

$$\Phi : x \mapsto \Phi(x).$$

That is not yet a complete physical field. Local values must be comparable, transportable, and dynamically related. A field therefore contains at least three structures:

$$\text{local distinction} + \text{relational transport} + \text{closure law}.$$

Unified Mechanics reads a field not as a collection of independent values placed inside a pre-existing container, but as the retained relational body whose update makes locations, particles, and measurable differences readable.

> ***The universe is not the place in which the field occurs. The universe is the field whose update makes place, action, and persistence readable.***

A particle can then be treated as a stable local equivalence class of field traversal. The particle is real, but its identity is conditional on the field preserving the distinctions by which it remains recoverable.

9.1 Vacuum is a physical environment

Laboratory isolation does not remove the field. It selects a controlled field environment. The electromagnetic vacuum has a mode structure, correlation functions, boundary conditions, and fluctuation spectrum. A free-space decay rate is therefore not a property of an atom considered without relation; it is the atom's transition rate relative to the mode structure called vacuum.

For an electric dipole transition in a general electromagnetic environment,

$$\Gamma(\mathbf{r}, \omega) = \frac{2\mu_0 \omega^2}{\hbar} \mathbf{d} \cdot \mathrm{Im}\mathbf{G}(\mathbf{r}, \mathbf{r}; \omega) \cdot \mathbf{d}^*.$$

The Green tensor $\mathbf{G}$ contains the entire environmental response: free propagation, material structure, reflection, absorption, resonance, and return. The decay probability belongs to the atom-field-environment relation.

10 Bounded Possibility

Consider water heated in an open saucepan. Liquid becomes vapour, the vapour enters the surrounding air, and its subsequent interactions disperse through a larger environment. Now consider the same transformation inside a sealed or strongly confining vessel. The material cannot simply leave. Pressure changes, vapour remains available to condense, heat can be added and removed, and the same material can repeatedly traverse different phases.

Open: liquid → vapour → air → dispersal

Bounded: liquid ⇄ vapour ⇄ wall ⇄ pressure ⇄ liquid

The difference is not merely that one system has a wall and the other does not. The boundary changes which relations can return. It keeps the products of a transformation available to participate in later transformations. The system becomes capable of changing state repeatedly rather than producing a one-way local loss.

> ***The container does not merely hold matter. It makes the transitions of that matter continue to belong to one body.***

Steam released into air does not stop interacting. It interacts with air molecules, temperature gradients, surfaces, humidity, and atmospheric motion. The important loss is not the number of microscopic relationships. The important loss is that those relationships cease to be recursively accountable to the original local system.

many interactions ≠ one retained relational history

An open system can therefore generate enormous relational activity while losing the ability to recover its own earlier states. The correlations are exported. The local system no longer has access to the same material, energy, and consequences required to close the cycle.

- **Relations spread into an enlarging environment.:** Relations repeatedly return through the same boundary conditions.
- **Material and energy may leave the local account.:** Material and energy remain more available to the local account.
- **Many paths become exits.:** Many paths become loops or revisitable transitions.
- **The original body loses recoverability.:** The body retains a shared history across state change.

A container should not be treated merely as another object placed around the system. Functionally, it is an operator on the system's transition structure. It suppresses some paths, redirects others, and creates returns that would not exist in the same form without confinement.

B : transition network → retained transition network

The boundary may reduce direct escape while increasing recurrence. In doing so, it changes not only where the system is located but what the system can repeatedly become. It can preserve phase cycling, pressure feedback, collision frequency, and delayed return. Constraint therefore changes the topology of possibility.

> ***A boundary removes dispersive paths so that transformative paths can remain available.***

In ordinary speech, possibility is often imagined as whatever could happen in the absence of restriction. Physical possibility is more demanding. A state is not physically possible for a particular system merely because it can be imagined. There must be an accessible path from the present condition to that state, and the resources required for the transition must remain available long enough for it to occur.

physical possibility = allowed + reachable + retainable

By preventing vapour from escaping, a vessel removes the path of immediate dispersal but preserves the paths of condensation, renewed evaporation, pressure equilibration, and repeated heat exchange. Constraint has removed one kind of freedom while creating a larger reusable family of local transformations.

This gives a direct reversal of the usual assumption: less freedom of escape can produce more freedom of transformation. Constraint does not necessarily make the possibility-space smaller. It may make the possibility-space more connected and more reusable.

A highly energetic, highly constrained system combines two operations that appear opposite only when they are placed on the same axis. Energy opens access to more transformations. Constraint requires those transformations to remain accountable to the same system.

energy → breadth of reachable possibility

constraint → continuity of probabilistic accountability

This is the sense in which a system can place the tightest leash on probability while placing the longest leash on possibility. The system is free to explore many states, but it is not free to pretend that each state belongs to a different universe of account. Conservation, boundary conditions, and prior interactions continue to weight what can actually occur.

> ***The richest production may occur when possibility is opened energetically while probability is forced to keep resolving inside one conserved relational whole.***

The phrase dense possibility does not mean an arbitrary number of imagined outcomes. It means a high concentration of reachable states connected by many available transitions, especially transitions that can return, recur, or feed into one another.

Let the possible system states form a directed graph G = (V, E). The vertices V are distinguishable states and the edges E are allowed transitions. An open dispersing system may contain extensive branching while having few return paths. A confined system can convert branches into cycles.

Dispersive: $S \to A \to A_1 \to$ outward loss

Recurrent: $S \to A \to B \to S$

Dense possibility is therefore not only a large number of states. It is a large number of mutually available transformations. Possibilities become structurally meaningful when reaching one state does not automatically destroy access to the rest of the state-space.

possibility density ≈ reachable states × transition connectivity × recurrence

The final expression is a heuristic definition, not yet a physical law. Its purpose is to expose the three ingredients that the ordinary word possibility tends to collapse together.

Probability becomes meaningful when the system persists long enough for its possibilities to be tested repeatedly under a continuing set of constraints. If the first transition destroys or exports the system, the alternatives function mainly as possible exits. When paths can be revisited, the system can establish stable relative weighting.

$S \rightleftarrows A,\ S \rightleftarrows B,\ S \rightleftarrows C,\ S \rightleftarrows D$

$p(A) + p(B) + p(C) + p(D) = 1$

The normalisation condition is not itself the new idea. The proposed interpretation is that a probability distribution becomes physically informative because its alternatives remain members of one retained account. Probability is not simply painted onto abstract possibility. It is the pattern that emerges when reachable possibilities repeatedly encounter the same body, invariants, and boundary conditions.

> ***Meaningful probability is possibility repeatedly tested inside persistent constraint.***

Meaning is used here in an operational rather than psychological sense. A distinction has physical meaning when it survives long enough, or leaves enough recoverable consequence, to affect a later distinction. A difference that is immediately erased without trace cannot participate in a continuing structure.

difference → interaction → retained consequence → possible return

difference → immediate dispersal → unrecoverable loss

The first sequence can accumulate history. The second cannot easily be distinguished from noise from the standpoint of the original system. Meaning therefore requires neither permanence nor a fixed object. It requires recoverability across transformation.

This connects directly to the semantic traversal system. Persistence does not mean that a thing remains unchanged. It means that enough relation survives the change for the distinction to be reconstructed, answered, or closed.

10.1 A graph-theoretic scaffold

Let a system's currently reachable state-space be represented by a directed graph

$$\mathcal{G}_C = (V_C, E_C),$$

where C denotes the active constraints, V_C the retained states, and E_C the admissible transitions.

An open dispersing system can possess many outward edges but few return cycles. A bounded system can remove escape edges while increasing recurrent connectivity. Define the recurrent edge set

$$E_C^{\circlearrowleft} = \{e \in E_C : e \text{ lies on at least one directed cycle}\}.$$

A provisional relational-density functional is

$$\eta(C) = \frac{|E_C^{\circlearrowleft}|}{|V_C|}.$$

The exact continuum generalisation remains domain-specific, but the conceptual distinction is precise: the number of abstract possibilities is not the same as the density of possibilities that remain mutually reachable within one retained whole.

Meaningful probability requires a normalised weighting over retained reachability:

$$p(v \mid C) = \frac{w_C(v)}{\sum_{u \in V_C} w_C(u)}.$$

The weights become physically meaningful only when the system persists long enough for its transition structure to constrain them.

11 The Body Before the Particle

The same reasoning changes how an isolated particle is interpreted inside a dense plasma. The particle may retain invariant labels such as charge, mass, and spin when analytically separated, but its current behaviour belongs to a larger relational body. Density, temperature, surrounding fields, and repeated collisions determine which local modes are accessible and how they are weighted.

particle alone ≠ particle-in-body

Removing a particle from the plasma can produce a cleaner object for measurement, but the act of removal also produces a new system. It eliminates the recurrence, pressure, collective fields, and state transitions that constituted the particle's role inside the plasma.

> ***Isolation may not reveal the object beneath the relations. Isolation may manufacture a new object by removing the relations that previously constituted it.***

The body may therefore be more primitive than the particle for some questions. A particle can remain real as a locally recoverable resolution without being the complete carrier of the field's behaviour. A droplet is real, but it does not contain the current of the river merely because both are water.

Quantum measurement is commonly narrated as a probe inspecting an object. In physical operation, the probe, target, apparatus, and environment transform one another. The record belongs to the closed chain of interaction rather than to the target alone.

probe → medium → interaction → redistribution → record

The present proposal adds a particular emphasis: the measuring relation should not be treated merely as a tool that reveals an already complete object. The relation may be the process through which probe, target, and outcome become separately recoverable identities.

Inside a dense body, attempting to isolate one particle may increasingly measure the boundary conditions of the isolation rather than an independent truth the particle carried before extraction. The useful question may shift from "What does this particle possess?" to "Which distinction remains recoverable across this complete interaction?"

The following definitions are deliberately provisional. They are offered to organise the idea before any claim of physical derivation.

- **Boundary B:** An operation that suppresses, redirects, or returns transitions so that consequences remain within a continuing system.
- **Possibility space ΩB:** The set of states reachable from the current state under the constraints and resources preserved by B.
- **Recurrence R:** The degree to which transitions return to, revisit, or remain causally available to the same system.
- **Distinguishability D:** The degree to which alternative states retain differences that can later be recovered.
- **Probability p:** A normalised weighting over reachable alternatives within one retained relational account.
- **Meaning M:** Recoverable difference accumulated across a traversal; not a subjective interpretation.
- **Object O:** A stable equivalence class of recurrent traversals that can be identified across change.

A heuristic meaning-capacity expression may then be written as:

$M_h \propto H(\Omega B) \cdot R \cdot D$

Here H(ΩB) represents the breadth of accessible possibility, R represents recurrence or retained return, and D represents distinguishability. The expression is not yet an empirical equation. It merely states the structural hypothesis that meaning collapses if possibility is absent, if nothing returns, or if all alternatives become indistinguishable.

A second useful quantity would compare recurrent transition weight with outward loss:

ρ = returned or retained transition weight / total transition weight

A system with high energetic access but very low ρ may explore many states while rapidly losing them from the local account. A system with high ρ but almost no energetic access may retain itself while exploring little. The proposed productive regime requires both accessible possibility and sufficient relational retention.

11.1 Dense media and quasiparticle identity

In a medium, the propagator of an excitation takes the schematic form

$$G(p) = \frac{1}{p^2 - m^2 - \Sigma(p; T, \mu) + i\epsilon}.$$

The self-energy Σ records the returned response of the surrounding body. Its real part shifts dispersion and effective mass; its imaginary part supplies damping and lifetime. The excitation is not a bare particle plus a later correction. It is the locally recoverable identity of a many-body relation.

This does not eliminate particle physics. It defines the conditions under which particle-like identity is physically available.

12 Gauge Fields as Preserved Comparability

Suppose a field $\psi(x)$ is described in local frames that can vary from point to point. The ordinary derivative compares values as though the local frames were already identical. A connection is required to preserve comparability across the distinction:

$$D_\mu = \partial_\mu + A_\mu.$$

In Unified Mechanics notation:

$$A_\mu = \text{the relational machinery required to transport identity across local distinction.}$$

The field strength is the residual nonclosure of two transport orders:

$$F_{\mu\nu} = [D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

For an Abelian connection the commutator vanishes. For a non-Abelian connection it records the order-sensitive relational remainder.

Curvature is therefore not merely a tensor assigned to a field. It is what remains when an identity transported through two logically distinct paths fails to return as the same local description:

$$F_{\mu\nu} = \text{remainder of non-closing relational traversal.}$$

Gauge transformations generate different local inscriptions of the same relational transport. PEA identifies descriptions that preserve gauge-invariant dynamics. PLA retains path ordering, holonomy, topology, and boundary data when they remain physically active.

13 Gravity as the Common Closure of All Fields

General relativity makes the relational structure used by every local measurement dynamical. The metric $g_{\mu\nu}$ defines intervals, proper time, null propagation, and causal comparison. The Levi-Civita connection transports spacetime identity; curvature records nonclosure.

The Einstein equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \, T_{\mu\nu}^{\mathrm{tot}}.$$

The total stress-energy contains the contributions of all matter and field sectors, including interaction terms:

$$T_{\mu\nu}^{\mathrm{tot}} = \sum_A T_{\mu\nu}^{(A)} + T_{\mu\nu}^{(\mathrm{int})}.$$

This gives a precise physical reading to the intuition that gravity is the sum of all fields. Gravity is not normally an arithmetic sum of gauge potentials. It is the common geometric response to the complete stress-energy account of every field. Each sector contributes to one shared closure structure.

Unified Mechanics therefore identifies gravity as

$$\boxed{\text{the geometric closure response of the total relational field account.}}$$

The metric is the machinery through which every local system participates in a common causal and observational whole. General relativity is the theory of that shared closure at classical geometric resolution.

Part IV — Quantum Action and State Space

14 Worked Derivation: Why a Quantum Transition Equation Has Its Form

Consider an initial state $|i\rangle$ and possible final states $|f\rangle$. The interaction Hamiltonian H_{int} relates them. First-order time-dependent perturbation theory gives

$$c_f(t) = -\frac{i}{\hbar}\int_0^t \langle f|H_{\text{int}}|i\rangle \exp\left(\frac{i(E_f - E_i)t'}{\hbar}\right) dt'.$$

Let

$$M_{fi} = \langle f|H_{\text{int}}|i\rangle, \qquad \Delta\omega_{fi} = \frac{E_f - E_i}{\hbar}.$$

Then

$$P_{i\to f}(t) = \frac{4\left|M_{fi}\right|^2}{\hbar^2}\frac{\sin^2\left(\Delta\omega_{fi}t/2\right)}{\Delta\omega_{fi}^2}.$$

Define the traversal kernel

$$\mathcal{T}_t(\Delta\omega) = \frac{4}{\hbar^2}\frac{\sin^2(\Delta\omega t/2)}{\Delta\omega^2}.$$

The probability factorises as

$$P_{i\to f}(t) = \underbrace{\left|M_{fi}\right|^2}_{\text{relational accessibility}} \underbrace{\mathcal{T}_t\left(\Delta\omega_{fi}\right)}_{\text{temporal traversal}}.$$

In the long-time rate limit,

$$\frac{1}{t}\mathcal{T}_t(\Delta\omega) \to \frac{2\pi}{\hbar}\delta\left(E_f - E_i\right).$$

Closing over the retained final possibilities produces Fermi's golden rule:

$$\Gamma = \frac{2\pi}{\hbar}\sum_f |\langle f|\, H_{\text{int}}|i\rangle|^2 \delta\left(E_f - E_i\right).$$

The equation can now be read as necessity rather than symbol inventory:

- **:** $i\rangle$
- **{:** $f\rangle$}
- H_{int}**:** Alternatives must be made reachable by relation.
- $\int_0^t d\,t'$**:** The relation must propagate through ordered time.
- **phase accumulation:** Compatible histories reinforce; incompatible histories cancel.
- $\delta\left(E_f - E_i\right)$**:** Persistent traversal resolves energetic accountability.
- $\sum_f$ **:** The complete retained possibility-space must close.
- Γ**:** The closed normal form is a transition rate.

The delta function is especially important. In the compressed formula it looks like a static constraint inserted from outside. In the expanded derivation it is the long-time residue of phase compatibility. The framework therefore explains why energy-compatible relations survive into a persistent rate.

15 Equivalent Action, Admissibility, and Superposition

Consider a retained initial condition I. The system admits a set of possible logical actions that begin from I:

$A(I) = \{ \alpha_1, \alpha_2, \ldots, \alpha_n \}$.

Not every imaginable action belongs to A(I). Each member must be allowed by the system's constraints, connected to the initial condition by an available relation, and capable of propagation. The set is therefore a physically distinguished possibility-space, not a catalogue of verbal possibilities.

15.0.1 No selection without a resolving distinction

Suppose α_1 and α_2 are both admissible and no physical event has yet distinguished one as the resolved record. Selecting α_1 at this stage would add information not supplied by the system. It would perform a resolution without a resolving operation. The system must therefore preserve both actions in its current description.

No resolution event ⇒ no admissible branch may be deleted.

This is the constitutive necessity behind superposition. The unresolved system cannot be represented by an ordinary exclusive choice, because exclusivity is itself a later physical result. It must be represented by a structure capable of retaining alternatives jointly while preserving their distinct amplitudes and phases.

15.0.2 Quotient by equivalent action

The raw history-space may contain many paths that are equivalent under the Principle of Equivalent Action. The physically meaningful possibility-space is therefore not simply A(I), but its quotient into equivalent-action classes:

$Q(I) = A(I) / \sim EA$.

Each basis label $|[\alpha]\rangle$ denotes not one microscopic path but a class of histories that realize the same resolved action. This is important: quantum alternatives need not be interpreted as a list of tiny classical objects. They can be interpreted as distinct classes of physically admissible execution.

15.0.3 The unresolved state

The system's unresolved state must retain every admissible class with its relational weight. The minimal expression is:

$|\Psi I\rangle = \Sigma[\alpha]\in Q(I)\ c[\alpha]\ |[\alpha]\rangle$.

The addition sign does not mean the system has already chosen several mutually exclusive records. It means the alternatives remain jointly available to the same closure operation. The coefficients are not classical probabilities at this stage. They are amplitudes carrying both magnitude and phase, because alternative logical actions can reinforce or cancel one another before resolution.

> ***The superposed state is a compressed ledger of unresolved action-classes, not a pile of simultaneously finished objects.***

Feynman's path formulation assigns a phase to each physical history according to its action. For a history α, the standard weighting is:

$w[\alpha] = \exp(i\ S[\alpha] / \hbar)$.

Under the new lens, this phase is the retained logical action of the history. The action functional is not merely a number attached after the path is known. It is the compressed accumulation of how the path traversed the system's relational constraints. The phase allows histories to remain distinguishable while still entering one shared closure.

The amplitude for one equivalent-action class is therefore a coherent sum over the histories inside that class:

$c[\alpha] = N\ \Sigma\beta\in[\alpha]\ \mu[\beta]\ \exp(i\ S[\beta] / \hbar)$, where $\mu[\beta]$ represents the appropriate measure or weighting and N normalizes the complete state. Equivalent Action groups histories by what they resolve into; Logical Action prevents their internal phase, order, and topology from being erased before the sum is performed.

15.0.4 Sequential and alternative composition

The algebra now acquires a physical explanation. Alternative actions combine additively because they are jointly retained possibilities entering one resolution. Sequential actions combine through composition, which appears multiplicatively at the amplitude level. Distributivity is then required because a prior action followed by either of two later alternatives must equal the combination of the two complete sequential histories:

$A \circ (B + C) = (A \circ B) + (A \circ C).$

This is the operational seed of linearity. A system that can distinguish alternatives, preserve them jointly, and compose them with earlier or later actions requires a representation in which alternative composition is additive and sequential composition distributes over it. The familiar vector-space form of quantum mechanics is the mature mathematical realization of that requirement.

If two unresolved action-classes lead to the same possible record, their amplitudes must be combined before probability is assigned:

$A(o) = c_1 + c_2.$

The resolved probability is then:

$P(o) = |c_1 + c_2|^2 = |c_1|^2 + |c_2|^2 + 2\,\mathrm{Re}(c_1\, c_2^*).$

The cross term is the mathematical trace of relation between the unresolved logical actions. A classical mixture removes this relation and retains only the separate weights:

$P_{mix}(o) = |c_1|^2 + |c_2|^2.$

This gives a precise distinction. Superposition is not merely uncertainty about which finished state already exists. It preserves cross-relations among admissible actions. A mixture has already lost, ignored, or physically dispersed those relations. In the language developed from the bounded-possibility argument, coherent constraint keeps the alternatives inside one accountable body; decoherence distributes their relational phase into an environment until the local alternatives can no longer return to one another as a coherent whole.

15.0.5 Superposition and meaningful probability

The earlier container argument can now be restated quantum mechanically. Constraint does not only delete possibilities. It can preserve return paths among them. When admissible alternatives remain phase-related inside one system, probability is not assigned independently to each path. It emerges from the complete relational closure of their amplitudes.

Meaningful quantum probability = retained possibility + phase relation + resolution + normalized closure.

The boundary conditions of an experiment, cavity, field, or worldsheet determine which actions remain available and how they return. High energetic accessibility opens possibilities; constraint keeps them accountable; coherent propagation allows them to interfere; closure converts the relational amplitude into an observable probability distribution.

Using the working semantic sequence developed in Unified Mechanics, the quantum state can be unfolded as follows.

Persist: An initial condition remains sufficiently identifiable to support a history and later comparison.

Distinguish: The system resolves a set of non-interchangeable admissible action-classes.

Relate: An interaction, Hamiltonian, boundary condition, or coupling makes particular transitions physically reachable.

Propagate: Each admissible action carries phase and ordered history through the system.

Recurse: Alternative histories, returns, loops, and repeated interactions remain jointly available to the unresolved account.

Resolve: Interference, conservation, compatibility, and measurement distinguish the combinations capable of producing a stable record.

Close: The complete outcome-space is normalized into an accountable probability distribution and recorded event.

Superposition occupies the middle of this traversal. It is the form taken by the system after admissible actions have been distinguished and related, but before their relational structure has been fully resolved and closed. This explains why superposition disappears neither by verbal decision nor by treating it as ignorance. It changes only when the physical relation among the alternatives changes.

16 From Admissible Logical Actions to Quantum State Space

The preceding chapter establishes the physical need to retain unresolved action classes. This chapter follows the mathematical closure required for that retention. The derivation is conditional on explicit operational commitments; its strength is that each quantum structure answers a necessity that appears earlier in the chain.

Let $\mathcal{A}$ be the set of physically admissible logical actions for a preparation. Each $a \in \mathcal{A}$ is an ordered history, not merely an endpoint.

PEA defines the quotient

$$\Omega = \mathcal{A}/\sim_{EA},$$

whose elements are equivalent-action classes

$$\alpha = [a]_{EA}.$$

PLA retains the trace map

$$\Lambda: \Omega \rightarrow \mathcal{T},$$

so that two classes can share an endpoint while remaining logically distinct whenever their ordered traces differ.

An unresolved preparation does not yet select one $\alpha \in \Omega$. Its first mathematical representation must therefore account for the complete admissible family

$$\Omega_P \subseteq \Omega.$$

This gives the first necessity:

No admissible action may be deleted before a physical resolution distinguishes it from the others.

16.0.1 Formal coherent combinations

Associate a basis symbol $|\alpha\rangle$ with each action class $\alpha \in \Omega_P$. The unresolved state is represented provisionally as

$$|\Psi\rangle = \sum_{\alpha \in \Omega_P} c_\alpha \, |\alpha\rangle.$$

At this stage, the coefficients c_α are not yet probabilities. They are relational weights whose algebra must be derived.

16.0.2 Why addition is required

If α and β remain unresolved, the representation must retain both:

$$|\alpha\rangle \oplus |\beta\rangle.$$

The operation must be associative and commutative for jointly retained alternatives:

$$(|\alpha\rangle + |\beta\rangle) + |\gamma\rangle = |\alpha\rangle + (|\beta\rangle + |\gamma\rangle),$$
$$|\alpha\rangle + |\beta\rangle = |\beta\rangle + |\alpha\rangle.$$

There must also be a null action 0 satisfying

$$|\alpha\rangle + 0 = |\alpha\rangle.$$

16.0.3 Why additive inverses are required

Quantum alternatives can cancel. A two-path arrangement can have a nonzero contribution from each path separately while their joint contribution vanishes. Therefore the algebra must contain an inverse contribution:

$$|\alpha\rangle + (-|\alpha\rangle) = 0.$$

Without additive inverses, destructive interference is impossible. Joint retention with possible cancellation therefore requires an abelian additive group.

16.0.4 Why scalar multiplication is required

Different admissible actions need not contribute equally. Introduce scalar weighting

$$c|\alpha\rangle.$$

Sequential physical composition must distribute over unresolved alternatives:

$$T(c_1|\alpha\rangle + c_2|\beta\rangle) = c_1T|\alpha\rangle + c_2T|\beta\rangle.$$

Likewise, scalar composition must satisfy

$$(c_1c_2)|\alpha\rangle = c_1(c_2|\alpha\rangle),$$

$$(c_1 + c_2)|\alpha\rangle = c_1|\alpha\rangle + c_2|\alpha\rangle.$$

The minimal algebraic closure is therefore a vector space over a scalar division algebra F.

Result 1 – Coherent-retention result. Joint retention, distributive propagation, and destructive cancellation force a linear state space before any specifically quantum notation is assumed.

16.0.5 Composition of action

For sequential histories a and b, ordinary action is additive:

$$S[a \circ b] = S[a] + S[b].$$

Let $w(S)$ be the scalar weight assigned to an action value. Logical composition requires

$$w(S_1 + S_2) = w(S_1)w(S_2).$$

For reversible unresolved propagation, the weight must not by itself change total accountability:

$$|w(S)| = 1.$$

Assume further that w is continuous and nontrivial. Then w is a continuous character of the real numbers under addition into $U(1)$. Every such character has the form

$$w(S) = e^{i\kappa S}$$

for a real constant κ.

Choosing the experimentally fixed action scale $\hbar$ gives

$$\boxed{w(S) = e^{iS/\hbar}}.$$

Changing orientation sends $\kappa \mapsto -\kappa$, giving the complex conjugate representation.

16.0.6 Why real positive weights are insufficient

If $w(S)$ were only a nonnegative real number, contributions could reinforce but could not continuously rotate into cancellation. The phase relation

$$e^{i\theta} + e^{i(\theta+\pi)} = 0$$

is the minimal continuous mechanism for complete destructive interference without deleting either action.

A two-dimensional real rotation can represent the same structure, but the complex number

$$e^{i\theta} = \cos\theta + i\sin\theta$$

is its minimal scalar compression. Complex amplitude is therefore not ornamental notation. It is the natural scalar representation of continuously composable relative action phase.

16.0.7 Action-state representation

The unresolved state may now be written

$$|\Psi\rangle = \sum_{\alpha} A_\alpha \, e^{iS_\alpha/\hbar} |\alpha\rangle,$$

or in a continuum of histories,

$$|\Psi\rangle \sim \int_{\mathcal{A}/\sim_{EA}} \mathcal{D}\,\alpha \;\; A[\alpha] e^{iS[\alpha]/\hbar} |\alpha\rangle.$$

PLA remains necessary because two histories may have equal action values while differing in order, topology, constraints, or boundary content.

Result 2 – Phase result. Additive action, sequential composition, continuity, reversibility, and interference force unit-modulus action weights of exponential complex form.

A state space is not physically useful unless two unresolved states can be compared. Introduce an overlap map

$$\langle \Phi | \Psi \rangle \in C.$$

The overlap must satisfy:

1. linearity in the retained state,
2. conjugate linearity in the comparison state,
3. conjugate symmetry,
4. positivity,
5. nondegeneracy.

Thus

$$\langle \Phi | a\Psi + b\Xi \rangle = a\langle \Phi | \Psi \rangle + b\langle \Phi | \Xi \rangle,$$

$$\langle a\Phi + b\Xi | \Psi \rangle = \overline{a}\langle \Phi | \Psi \rangle + \overline{b}\langle \Xi | \Psi \rangle,$$

$$\langle \Phi | \Psi \rangle = \overline{\langle \Psi | \Phi \rangle},$$

$$\langle \Psi | \Psi \rangle \geq 0, \qquad \langle \Psi | \Psi \rangle = 0 \Leftrightarrow |\Psi\rangle = 0.$$

The norm is

$$\parallel \Psi \parallel = \sqrt{\langle \Psi | \Psi \rangle}.$$

If physically convergent sequences of preparations are required to have limits inside the theory, the inner-product space is completed. The result is a complex Hilbert space $\mathcal{H}$.

Orthogonality

$$\langle \alpha | \beta \rangle = 0$$

represents perfectly distinguishable resolutions. Nonzero overlap represents unresolved relational compatibility.

Result 3 – Hilbert-completion result. Linear coherent retention plus positive continuous relational comparison produces a pre-Hilbert space; closure under physical limits produces a Hilbert space.

Let $\{|i\rangle\}$ be a complete set of mutually exclusive sharp resolutions. Expand

$$|\Psi\rangle = \sum_i c_i \, |i\rangle.$$

A closed preparation must resolve somewhere inside its complete outcome account:

$$\sum_i p_i = 1.$$

The state norm is therefore fixed to unity:

$$\langle \Psi | \Psi \rangle = 1.$$

For an orthonormal basis,

$$\langle\Psi|\Psi\rangle = \sum_i |c_i|^2 = 1.$$

Normalization is not an arbitrary tidying rule. It is the mathematical closure condition that no probability is lost outside the system's declared resolution space.

16.0.8 Resolution effects

Represent an outcome by an effect E satisfying

$$0 \leq E \leq I.$$

A complete measurement is a family $\{E_i\}$ with

$$\sum_i E_i = I.$$

Let $\mu_\rho(E)$ be the probability assigned to effect E in preparation ρ.

16.0.9 The Equivalent-Resolution requirement

If the same physical effect E occurs inside two different complete measurement arrangements, PEA requires the same probability assignment. The outcome's probability cannot depend on an irrelevant redescription of the larger measurement context:

$$E \sim_{EA} E' \quad \Rightarrow \quad \mu_\rho(E) = \mu_\rho(E').$$

For jointly exclusive effects,

$$E + F \leq I \quad \Rightarrow \quad \mu_\rho(E + F) = \mu_\rho(E) + \mu_\rho(F).$$

Also,

$$\mu_\rho(I) = 1.$$

Gleason's theorem, and its POVM extensions due to Busch and Caves–Fuchs–Manne–Renes, imply that such an additive noncontextual probability measure has the trace form

$$\boxed{\mu_\rho(E) = Tr(\rho E)},$$

where

$$\rho \geq 0, \qquad Tr\rho = 1.$$

For a pure state

$$\rho = |\Psi\rangle\langle\Psi|$$

and a sharp outcome

$$P_i = |i\rangle\langle i|,$$

we obtain

$$p_i = Tr[(|\Psi\rangle\langle\Psi|)(|i\rangle\langle i|)] = |\langle i|\Psi\rangle|^2.$$

Thus

$$\boxed{p_i = |c_i|^2}.$$

The original Gleason theorem applies to projective measurements in Hilbert-space dimension at least three. The POVM versions recover the same trace rule for two-dimensional systems under the generalized-effect assumptions.

Result 4 – Born-closure result. Once equivalent physical effects must retain the same additive probability across every complete resolution context, the state is represented by a density operator and probabilities take the Born trace form.

A coherent superposition is

$$|\Psi\rangle = c_1|1\rangle + c_2|2\rangle.$$

Its density operator is

$$\rho_\Psi = |c_1|^2|1\rangle\langle 1| + |c_2|^2|2\rangle\langle 2| + c_1\overline{c_2}|1\rangle\langle 2| + c_2\overline{c_1}|2\rangle\langle 1|.$$

The off-diagonal terms retain the relational consequences between unresolved actions.

A classical mixture is

$$\rho_{mix} = p_1|1\rangle\langle 1| + p_2|2\rangle\langle 2|.$$

It contains no coherent cross-action terms.

Therefore:

$$\boxed{\text{superposition} = \text{jointly retained alternatives whose relations remain physically consequential}}$$

while

$$\boxed{\text{mixture} = \text{a classical distribution over alternatives whose relative logical action has been lost or ignored.}}$$

Decoherence is the transfer of these cross-action relations into a larger environment, not the retroactive claim that only one alternative had existed.

Let a closed unresolved state propagate by a map

$$|\Psi\rangle \mapsto T|\Psi\rangle.$$

Closed propagation must preserve every future transition probability:

$$|\langle\Phi|\Psi\rangle|^2 = |\langle T\Phi|T\Psi\rangle|^2.$$

Wigner's theorem states that a bijection on rays preserving transition probabilities is implemented by either a unitary or antiunitary operator [3].

Time propagation satisfies

$$U(0) = I, \qquad U(t+s) = U(t)U(s),$$

and is continuously connected to the identity. An antiunitary transformation is not continuously connected to I within such a one-parameter evolution. Therefore ordinary continuous time propagation is unitary:

$$U(t)^\dagger U(t) = I.$$

Hence

$$\| U(t)\Psi \|^2 = \| \Psi \|^2,$$

so total probability remains closed.

Result 5 – Unitary-propagation result. Reversible propagation preserving all relational resolution probabilities is unitary when the propagation varies continuously from the identity.

Assume $U(t)$ is a strongly continuous one-parameter unitary group. Stone's theorem gives a unique self-adjoint generator H such that

$$\boxed{U(t) = e^{-iHt/\hbar}}.$$

The group property

$$U(t+s) = U(t)U(s)$$

is the mathematical expression that sequential temporal traversals compose.

For

$$|\Psi(t)\rangle = U(t)|\Psi(0)\rangle,$$

differentiate:

$$\frac{d}{dt}|\Psi(t)\rangle = -\frac{i}{\hbar}H|\Psi(t)\rangle.$$

Therefore

$$\boxed{i\hbar\frac{d}{dt}|\Psi(t)\rangle = H|\Psi(t)\rangle.}$$

Self-adjointness of H ensures real spectral values and unitary propagation.

The Schrödinger equation is therefore the infinitesimal normal form of continuous closed logical action.

A sharp observable with possible resolved values a_i is represented by

$$A = \sum_i a_i\, P_i,$$

where

$$P_i P_j = \delta_{ij} P_i, \qquad \sum_i P_i = I.$$

The expected resolved value is

$$\langle A\rangle_\rho = Tr(\rho A).$$

General measurements are POVMs $\{E_m\}$:

$$E_m \geq 0, \qquad \sum_m E_m = I,$$

with

$$p(m) = Tr(\rho E_m).$$

A measurement instrument resolves both probability and post-resolution state. In Kraus form,

$$\mathcal{I}_m(\rho) = \sum_\alpha M_{m\alpha}\, \rho M_{m\alpha}^\dagger,$$

$$E_m = \sum_\alpha M_{m\alpha}^\dagger\, M_{m\alpha},$$

$$p(m) = Tr[\mathcal{I}_m(\rho)],$$

$$\rho_m = \frac{\mathcal{I}_m(\rho)}{p(m)}.$$

The completeness condition

$$\sum_{m,\alpha} M_{m\alpha}^\dagger\, M_{m\alpha} = I$$

is again closure: the complete instrument loses no probability from its declared outcome space.

Let systems A and B have action-state spaces $\mathcal{H}_A$ and $\mathcal{H}_B$. Independent preparations must compose bilinearly:

$$(a|\psi_1\rangle + b|\psi_2\rangle) \boxtimes |\phi\rangle = a(|\psi_1\rangle \boxtimes |\phi\rangle) + b(|\psi_2\rangle \boxtimes |\phi\rangle),$$

$$|\psi\rangle \boxtimes (c|\phi_1\rangle + d|\phi_2\rangle) = c(|\psi\rangle \boxtimes |\phi_1\rangle) + d(|\psi\rangle \boxtimes |\phi_2\rangle).$$

The tensor product is the universal vector space representing bilinear composition:

$$\boxed{\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B}.$$

Product actions are

$$|\Psi_{AB}\rangle = |\psi_A\rangle \otimes |\phi_B\rangle.$$

A general joint action is

$$|\Psi_{AB}\rangle = \sum_{ij} c_{ij} \, |i\rangle_A \otimes |j\rangle_B.$$

It is entangled when no factorization exists:

$$|\Psi_{AB}\rangle \neq |\psi_A\rangle \otimes |\phi_B\rangle.$$

Thus entanglement is not a mysterious extra bond between finished particles. It is a joint logical action whose complete state cannot be decomposed into independent subsystem actions.

Local tomography – the requirement that a composite state be determined by correlations among local measurements – is a major reconstruction principle that selects ordinary complex quantum theory over standard real and quaternionic alternatives in broad operational settings [1,2,8].

A preparation procedure may randomly produce pure states $|\psi_k\rangle$ with classical weights q_k:

$$\rho = \sum_k q_k \, |\psi_k\rangle\langle\psi_k|, \qquad q_k \geq 0, \qquad \sum_k q_k = 1.$$

Different ensembles can generate the same density operator:

$$\{q_k, |\psi_k\rangle\} \neq \{r_j, |\phi_j\rangle\}$$

while

$$\sum_k q_k \, |\psi_k\rangle\langle\psi_k| = \sum_j r_j \, |\phi_j\rangle\langle\phi_j|.$$

They then give identical probabilities for every effect E:

$$Tr(\rho E)$$

is the same for both preparations. Therefore the density operator is an Equivalent-Action class of preparation procedures relative to all possible observations.

PLA can still distinguish the actual preparation history when that history later becomes physically accessible through an enlarged experiment.

For propagation from x_i to x_f, the path integral is

$$K(x_f, t_f; x_i, t_i) = \int_{x_i}^{x_f} \mathcal{D}\, x(t) \, \exp\left(\frac{i}{\hbar} S[x]\right).$$

This is the continuum realization of the action-state construction:

- the paths are admissible logical actions;
- the action phase is their compositional scalar weight;
- addition retains them jointly;
- interference compares their relative action;
- the final propagator closes them into one transition amplitude.

In the semiclassical limit, expand around a stationary path x_{cl}:

$$S[x_{cl} + \eta] = S[x_{cl}] + \delta S[x_{cl}; \eta] + \frac{1}{2} \delta^2 S[x_{cl}; \eta] + \cdots.$$

When

$$\delta S[x_{cl}] = 0,$$

nearby phases vary least rapidly and reinforce. Away from stationary action, phases cancel under integration. Thus the classical equation is the stable closure of quantum logical actions:

$$\boxed{\delta S = 0.}$$

This supplies a direct bridge between the Principle of Logical Action, quantum superposition, and classical variational physics.

The complete derivation can be compressed into the following sequence of necessity classes.

16.0.10 Coherent retention

Semantic requirement:

Retain every unresolved admissible action.

Mathematical closure:

$$|\Psi\rangle = \sum_i c_i \, |i\rangle.$$

16.0.11 Interference-capable scalar structure

Semantic requirement:

Permit constructive and destructive relation while preserving every admissible action.

Mathematical closure:

$$c_i \in C, \qquad w(S) = e^{iS/\hbar}.$$

16.0.12 Relational comparison

Semantic requirement:

Compare action states without erasing coherent combination.

Mathematical closure:

$$\langle \Phi | \Psi \rangle, \qquad \parallel \Psi \parallel^2 = \langle \Psi | \Psi \rangle.$$

16.0.13 Complete outcome accountability

Semantic requirement:

Close the entire declared resolution space without probability loss.

Mathematical closure:

$$\langle \Psi | \Psi \rangle = 1, \qquad \sum_i E_i = I.$$

16.0.14 Equivalent resolution

Semantic requirement:

Assign the same additive probability to the same physical effect in every equivalent context.

Mathematical closure:

$$p(E) = Tr(\rho E).$$

16.0.15 Reversible propagation

Semantic requirement:

Preserve every future transition probability through closed evolution.

Mathematical closure:

$$U(t)^{\dagger}U(t) = I, \qquad U(t+s) = U(t)U(s),$$
$$U(t) = e^{-iHt/\hbar}, \qquad i\hbar\dot{\Psi} = H\Psi.$$

16.0.16 Composite action

Semantic requirement:

Compose independent systems bilinearly and retain joint actions that cannot be separated.

Mathematical closure:

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B,$$
$$|\Psi_{AB}\rangle \neq |\psi_A\rangle \otimes |\phi_B\rangle$$

for an entangled state.

The following classes allow the conventional mathematics to be written once and subsequently compressed without losing the expansion path.

16.0.17 Class Q0 – Admissibility

Expanded form

$$\Omega_P = \{[a]_{EA} : a \text{ is physically admissible under } P\}.$$

Compressed sentence

Rationality retains the complete equivalent-action space permitted by the preparation.

16.0.18 Class Q1 – Coherent retention

Expanded form

$$|\Psi\rangle = \sum_{\alpha \in \Omega_P} c_\alpha \, |\alpha\rangle.$$

Compressed sentence

Unresolved admissible actions remain jointly retained.

16.0.19 Class Q2 – Action phase

Expanded form

$$w(S_1 + S_2) = w(S_1)w(S_2), \qquad |w(S)| = 1, \qquad w(S) = e^{iS/\hbar}.$$

Compressed sentence

Sequential logical action composes as relative phase.

16.0.20 Class Q3 – Relational overlap

Expanded form

$$\langle \Phi | \Psi \rangle, \qquad \| \Psi \|^2 = \langle \Psi | \Psi \rangle.$$

Compressed sentence

Action states are compared by preserved relational overlap.

16.0.21 Class Q4 – Closed probability

Expanded form

$$p(E) = Tr(\rho E), \qquad \sum_i E_i = I.$$

Compressed sentence

Equivalent resolutions receive one additive, normalized probability account.

16.0.22 Class Q5 – Closed propagation

Expanded form

$$U(t)^\dagger U(t) = I, \qquad U(t+s) = U(t)U(s), \qquad U(t) = e^{-iHt/\hbar}.$$

Compressed sentence

Unresolved action propagates reversibly while preserving every future resolution.

16.0.23 Class Q6 – Composition

Expanded form

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B.$$

Compressed sentence

Independent action spaces compose bilinearly; irreducible joint action is entanglement.

16.0.24 Class Q7 – Classical closure

Expanded form

$$K = \int \mathcal{D}\gamma\, e^{iS[\gamma]/\hbar}, \qquad \delta S = 0 \text{ in the stationary-phase limit.}$$

Compressed sentence

Quantum logical actions interfere until stationary equivalent action survives as classical form.

16.1 The quantum-state normal form

The chain can be compressed as

$$\boxed{\text{admissibility} \to \text{coherent retention} \to \text{complex phase} \to \text{inner product} \to \text{normalisation} \to \text{unitary propagation} \to \text{Born resolution}}$$

A quantum state is therefore

$$\boxed{\text{the normalised relational account of unresolved admissible logical actions.}}$$

Quantum mechanics is not added to the system after systemhood. It is the mathematical structure required when a system must preserve unresolved, mutually consequential alternatives without deleting relations that can still affect closure.

Part V — Strings and Geometric Closure

17 String Theory as Extended Logical Action

Point-particle quantum mechanics sums histories of worldlines. String theory replaces the worldline with a worldsheet. The action is no longer the traversal of one point through spacetime, but the propagation of an extended relation whose local and global structure both matter.

$Z_{string} = \Sigma_{topologies} \int DX\, Dh / (\mathrm{Diff} \times \mathrm{Weyl}) \cdot \exp(i\, S_P[X,h] / \hbar)$.

This expression is the string-theoretic realization of unresolved logical action. The alternatives are embeddings X, worldsheet metrics h, gauge-equivalent descriptions, moduli, and topologically distinct surfaces. The quantum state retains them before final scattering data is resolved. The string does not merely add more particle states. It geometrizes the logical action itself.

17.0.1 Nambu–Goto and Polyakov as Equivalent Action

$S_{NG} = -T \int d^2\sigma \sqrt{(-\det \gamma_{ab})}$

$S_P = -(T/2) \int d^2\sigma \sqrt{(-h)}\, h^{ab}\, G_{\mu\nu}(X)\, \partial_a X^\mu\, \partial_b X^\nu$

The Nambu–Goto action directly measures worldsheet area. The Polyakov action introduces an independent worldsheet metric and additional gauge structure. Classically, resolving the auxiliary metric recovers the same string equations. The formulations are therefore Equivalent Action representations. They are not automatically identical Logical Actions, because their internal fields, constraints, gauge reductions, measures, and quantization procedures are different.

17.0.2 Superposition of topologies

String interaction is encoded globally by splitting, joining, and handles of the worldsheet. Locally similar action densities can belong to globally different logical actions. A cylindrical propagation, a pair-of-pants splitting, and a handled loop may share the same local field law while representing different global executions. The Principle of Logical Action therefore requires topology to remain part of the action trace until the amplitude is closed.

Same local density ≠ same global logical action.

Superposition in string theory is consequently deeper than a particle occupying several eigenstates. It is the unresolved retention of extended geometrical histories, including histories whose distinction lies in global connection rather than local material content.

18 Nambu–Goto and Polyakov under PEA and PLA

A relativistic string admits the Nambu–Goto action

$$S_{\mathrm{NG}} = -T \int d^2\sigma \sqrt{-\det \gamma_{ab}},$$

where

$$\gamma_{ab} = G_{\mu\nu}(X)\, \partial_a X^\mu\, \partial_b X^\nu$$

is the induced worldsheet metric.

The Polyakov action is

$$S_{\mathrm{P}} = -\frac{T}{2} \int d^2\sigma \sqrt{-h}\, h^{ab} G_{\mu\nu}(X)\, \partial_a X^\mu\, \partial_b X^\nu.$$

Varying the auxiliary metric h_{ab} gives the vanishing worldsheet stress tensor. Resolving that constraint makes h_{ab} conformal to the induced metric, recovering the Nambu–Goto dynamics. Thus

$$S_{\mathrm{NG}} \sim_{EA} S_{\mathrm{P}}.$$

PEA identifies one classical string action class. PLA distinguishes their internal machinery:

$$\Lambda(S_{\mathrm{NG}}) \neq \Lambda(S_{\mathrm{P}})$$

before all auxiliary, gauge, and quantum structures are resolved. The Polyakov formulation introduces an independent worldsheet metric, diffeomorphism symmetry, Weyl symmetry, gauge fixing, ghost measure, and anomaly conditions. These are not visible in the Nambu–Goto endpoint statement of worldsheet area.

The example proves the need for both principles. Equivalent Action says the classical embeddings belong to one invariant class. Logical Action says the routes through that class cannot be declared identical before their constraints, measures, and quantum closures have been compared.

19 Worldsheet Superposition and Topology

A point particle carries a worldline $X^\mu(\tau)$. A string carries a worldsheet $X^\mu(\tau,\sigma)$. Its logical action is therefore extended: propagation and interaction are encoded in the geometry and topology of a two-dimensional surface.

The perturbative string partition function is schematically

$$Z = \sum_{\Sigma} g_s^{-\chi(\Sigma)} \int \frac{\mathcal{D}X\,\mathcal{D}h}{\mathrm{Diff}\times\mathrm{Weyl}} \exp\left(\frac{i}{\hbar} S_{\mathrm{P}}[X,h]\right).$$

The sum is over admissible worldsheet topologies Σ. Quantum mechanics supplies the coherent weighting of unresolved actions. String theory extends those actions into surfaces whose global topology carries interaction history.

A local worldsheet density can be the same on a cylinder, a splitting surface, or a higher-genus surface. The global logical action is different. PLA therefore requires topology to remain in $\Lambda(S)$ until the full amplitude closes.

20 From Worldsheet Consistency to Spacetime Gravity

A string propagating through background fields is described by a nonlinear sigma model. A representative bosonic action is

$$S_\sigma = -\frac{1}{4\pi\alpha'}\int d^2\sigma\sqrt{-h}\left[h^{ab}G_{\mu\nu}(X)\,\partial_a X^\mu\,\partial_b X^\nu + \epsilon^{ab}B_{\mu\nu}(X)\,\partial_a X^\mu\,\partial_b X^\nu + \alpha' R^{(2)}\Phi(X)\right].$$

Quantum consistency requires the worldsheet Weyl anomaly to vanish. The beta functions of the background fields must therefore satisfy

$$\beta^G_{\mu\nu} = 0, \qquad \beta^B_{\mu\nu} = 0, \qquad \beta^\Phi = 0.$$

At leading order,

$$\beta^G_{\mu\nu} = \alpha'\left(R_{\mu\nu} + 2\nabla_\mu\nabla_\nu\Phi - \frac{1}{4}H_{\mu\rho\sigma}H_\nu{}^{\rho\sigma}\right) + O(\alpha'^2).$$

The spacetime fields must therefore obey gravitational field equations so that the quantum worldsheet can close consistently.

This is the central three-theory bridge:

$\boxed{\text{quantum admissibility} \to \text{extended worldsheet action} \to \text{consistency constraint} \to \text{spacetime geometric closure.}}$

General relativity appears not as an unrelated classical theory attached afterward, but as the low-energy geometric closure of quantum extended action.

21 The Graviton as a Closed Relational Mode

The first excited closed-string state contains a left-moving and right-moving oscillator:

$$\alpha_{-1}^{\mu}\tilde{\alpha}_{-1}^{\nu}|k\rangle.$$

Its tensor product decomposes into a symmetric traceless field, an antisymmetric field, and a scalar trace:

$$\alpha_{-1}^{\mu}\tilde{\alpha}_{-1}^{\nu} \to G_{\mu\nu} \oplus B_{\mu\nu} \oplus \Phi.$$

The symmetric massless spin-two component is the graviton mode. Under the Unified Mechanics lens, it is the resolved symmetric relational mode of a closed left-right action. The structure again displays one whole resolving into direct, crossed, and retained relational sectors without ceasing to be one closed string state.

22 Quantum Mechanics, String Theory, and General Relativity

The three theories can now be assigned distinct roles in one operation:

- **Quantum mechanics:** Retains and weights unresolved admissible actions.
- **String theory:** Realises those actions as extended worldsheets with topology and left-right structure.
- **General relativity:** Supplies the dynamical spacetime geometry required for the extended quantum action to close.
- **Unified Mechanics:** Denotes the invariant logical grammar and equivalence machinery shared by all three.

The bridge is

$$\text{Possibility} \xrightarrow{\text{QM}} \text{weighted histories} \xrightarrow{\text{string}} \text{extended action surfaces} \xrightarrow{\text{consistency}} \text{spacetime equations} \xrightarrow{\text{GR}} \text{dynamical relational geometr}$$

The framework does not create a new physical mechanism beside the three theories. It identifies why their mathematical structures occupy successive stages of one system's action.

Part VI — Observation, Machinery, and Retained Reality

23 The Principle of Observation

The prior action-based reconstruction identified a quantum state as a normalised relational account of unresolved admissible logical actions. In a basis of mutually distinguishable alternatives,

$|\psi_S\rangle = \Sigma_i c_i |s_i\rangle, \Sigma_i |c_i|^2 = 1$. (1)

Equation (1) preserves every admissible alternative and the phase relations between them. It therefore describes an unresolved physical account, not an observed record. The next question is not merely how a probability is computed, but how an unresolved action-space becomes a persistent distinction in another physical system.

Operational hinge Superposition specifies what must remain jointly retained before resolution. Observation specifies how one system becomes a retained account of another. Mechanical observation specifies the machinery that performs this conversion.

The complete bridge proposed here is

Admissibility → Superposition → Mechanical distinction → Correlation → Propagation → Stabilisation → Retained record → Semantic closure. (2)

Principle of Observation (PO) An observation occurs whenever a distinction in one system becomes physically related to and persistently retained by another system. Consciousness is not required at this level. Observation is the production of an observer-relative fact through retained physical correlation.

$\mathfrak{O}_{O\leftarrow S} = \mathcal{C}_O \circ \mathcal{P}_O \circ \mathcal{R}_{OS} \circ \mathcal{D}_O(S)$.

Here S denotes the observed system and O denotes the observer or reference system. The components are

- **$\mathcal{D}_O(S)$ — Distinction:** A target-relative difference becomes available to O.
- **$\mathcal{R}_{OS}$ — Relation:** S and O physically interact or share a preparation correlation.
- **$\mathcal{P}_O$ — Persistence:** The difference leaves a stable enough change in O to matter later.
- **$\mathcal{C}_O$ — Closure:** The retained difference becomes an attributable observer-relative fact.

At this level, an atom can observe another atom in the relational sense: an interaction creates a correlation that changes the future behaviour of both. This agrees with relational approaches in which physical facts are relative to systems and no fundamental human observer-observed division is imposed. [1]

23.0.1 Minimal relational observation

$\Sigma_i c_i |s_i\rangle|o_0\rangle \to \Sigma_i c_i |s_i\rangle|o_i\rangle$. (3)

The observer states $|o_i\rangle$ now contain distinctions correlated with the alternatives $|s_i\rangle$. Equation (3) is already observation under PO because the relation has changed what is physically attributable relative to O. It is not yet necessarily a macroscopic measurement, a classical record, or a conscious experience.

24 The Principle of Mechanical Observation

Principle of Mechanical Observation (PMO) A mechanical observation is an ordered physical process in which the observer has an internal architecture that selects admissible distinctions, couples them to distinguishable channels, propagates and stabilises the resulting signal, and retains it as an identifiable record. Mechanical means implemented by a mechanism, not necessarily artificial.

$\mathfrak{O}^{mech}(O \leftarrow S) = \mathcal{C}_O \circ \mathcal{R}ec_O \circ \mathcal{A}_O \circ \mathcal{P}roj_O \circ \mathcal{U}\{SM\}.$

The observer machinery is represented as the tuple

$\boldsymbol{\mathcal{M}}_O = (\mathcal{H}_M, \sigma_0, U_{SM}, \{\Pi_k\}, \mathcal{A}, \mathcal{R}ec)$. (4)

Its structure is

- $\mathcal{H}_M$: the state-space of the observing mechanism.
- σ_0: its prepared ready state.
- U_{SM}: the system-machinery coupling.
- $\{\Pi_k\}$: the distinguishable output or pointer channels.
- $\mathcal{A}$: amplification, redundancy, or stabilisation.
- $\mathcal{R}ec$: production of a persistent record.

Subclass relation Every mechanical observation is an observation, but not every observation is mechanical in the structured sense: PMO ⇒ PO, while PO does not imply PMO.

24.0.1 Why the structure is genuinely different

A minimal observer only needs to retain a correlation. A mechanical observer must impose an ordered channel architecture. It therefore carries additional internal distinctions: ready state versus triggered state, one output channel versus another, signal versus noise, transient response versus persistent record, and record versus interpretation.

PO: S ↔ O → retained difference. PMO: S → selector → transducer → channel → amplifier → record. (5)

The machinery is not a passive window onto a fully prewritten classical property. It defines the physical question enacted on the system. In general quantum measurement theory, a measurement is represented by effects $\{E_k^{(\boldsymbol{\mathcal{M}})}\}$ satisfying

$E_k^{(\boldsymbol{\mathcal{M}})} \geq 0, \ \Sigma_k E_k^{(\boldsymbol{\mathcal{M}})} = I$, (6)

$p(k \mid \rho, \boldsymbol{\mathcal{M}}) = \mathrm{Tr}(\rho E_k^{(\boldsymbol{\mathcal{M}})})$. (7)

The same incoming state ρ can therefore yield different probability distributions under different observer architectures $\boldsymbol{\mathcal{M}}$. The apparatus does not create arbitrary outcomes: the state and the interaction constrain them. But the apparatus determines which distinction is made mechanically available as a record.

24.0.2 Qubit example

For a spin-1/2 state with Bloch vector r, a measurement along a unit direction n has effects

$\Pi_\pm^{(n)} = \tfrac{1}{2}(I \pm n\cdot\sigma)$, (8)

$p_\pm(n) = \mathrm{Tr}(\rho \Pi_\pm^{(n)}) = \tfrac{1}{2}(1 \pm r\cdot n)$. (9)

Rotating the measuring magnet changes n and therefore changes the distinction being physically enacted. Observer architecture is thus part of the complete state-to-record operation.

Let the system enter the observation in a coherent superposition and let the apparatus begin in a ready state $|M_0\rangle$:

$|\psi_S\rangle|M_0\rangle = (\Sigma_i c_i|s_i\rangle)|M_0\rangle$. (10)

An ideal channel-defining interaction satisfies

$$U_{SM}(|s_i\rangle|M_0\rangle) = |s_i\rangle|M_i\rangle. \quad (11)$$

By linearity,

$$U_{SM}[(\Sigma_i c_i|s_i\rangle)|M_0\rangle] = \Sigma_i c_i|s_i\rangle|M_i\rangle. \quad (12)$$

Transfer proposition The first stage of observation does not erase superposition. It transfers the unresolved action from the target alone into a larger correlated system-apparatus structure.

This is the exact transition from superposition to observer relation. Before coupling, the alternatives are carried by S. After coupling, they are carried by correlations between S and M. The observer machinery has made the alternatives mechanically discriminable, but it has not yet by unitary evolution alone produced one uniquely selected global branch.

Real apparatus records interact with an environment E. If the environment begins in $|E_0\rangle$ and becomes correlated with the pointer alternatives,

$$\Sigma_i c_i|s_i\rangle|M_i\rangle|E_0\rangle \to \Sigma_i c_i|s_i\rangle|M_i\rangle|E_i\rangle. \quad (13)$$

The reduced system-apparatus density operator is

$$\rho_{SM} = \Sigma_{i,j} c_i c_j^* \gamma_{ij} |s_iM_i\rangle\langle s_jM_j|, \ \gamma_{ij}=\langle E_j|E_i\rangle. \quad (14)$$

When distinct records leave nearly orthogonal environmental traces, $\gamma_{ij} \approx 0$ for $i \neq j$, and locally

$$\rho_{SM} \approx \Sigma_i |c_i|^2 |s_iM_i\rangle\langle s_iM_i|. \quad (15)$$

Environment-induced decoherence explains why certain pointer records are stable and why interference between macroscopically distinct records becomes locally inaccessible. [2]

Record stability condition A candidate record is mechanically real for an observer when it remains distinguishable and recoverable through the subsequent interactions that constitute the observer's memory or readout chain.

$\text{Record}(k) \Leftrightarrow \text{Dist}(M_k,M_j)$ persists under the relevant future dynamics for all $j \neq k$.

This formulation deliberately separates stable record production from any one metaphysical interpretation of global outcome selection. Decoherence provides the mechanical stability and effective noninterference of records; interpretations of quantum mechanics disagree about the global ontology of the remaining branches. PMO identifies the common physical operation that every interpretation must still represent.

The standard mathematical object closest to PMO is a quantum instrument $\{\mathcal{I}_k\}$. It encodes both the probability of each record and the state transformation conditioned on that record:

$$p_k = \mathrm{Tr}[\mathcal{I}_k(\rho)], \quad (16)$$

$$\rho_{S|k} = \mathcal{I}_k(\rho) / p_k, \quad (17)$$

$$\Sigma_k \mathrm{Tr}[\mathcal{I}_k(\rho)] = 1. \quad (18)$$

Completely positive instruments can be represented through indirect measurement models involving a physical apparatus, a system-apparatus interaction, and an apparatus readout. This is precisely the mathematical content required by mechanical observation: an abstract state-to-outcome map must be physically realisable as machinery. [3]

24.0.3 Explicit apparatus dilation

For apparatus state σ_0, coupling U_{SM}, and pointer projector Π_k, an idealised instrument can be written

$$\mathcal{I}_k(\rho) = \mathrm{Tr}_M[(I\otimes\Pi_k) U_{SM}(\rho\otimes\sigma_0)U_{SM}^\dagger (I\otimes\Pi_k)]. \quad (19)$$

Equation (19) is a complete mechanical-observation sentence in conventional mathematics: prepare machinery, couple system and machinery, distinguish pointer channel k, discard inaccessible apparatus detail, and retain the conditional system state.

25 Human Reflexive Observation

A human is not required for PO or for the mechanical production of a record. The specifically human scientific distinction appears when the observer can model the target, model the apparatus, re-enter the observation as a new object of observation, and communicate the result semantically.

$\mathfrak{D}_H^{(2)}(S) = \Lambda_H \circ \mathrm{Rec}_H \circ \mathfrak{D}_{H\leftarrow S}^{mech}$. (20)

Here Rec_H denotes reflexive re-entry and Λ_H denotes semantic denotation into language, diagrams, or mathematics. The human observer can ask why a basis was selected, what the detector excluded, how the signal was amplified, what assumptions entered the reconstruction, and whether a different observer architecture would yield another legitimate distinction.

Reflexive mechanical observer A mechanical observer whose internal records can themselves become targets of further observation, comparison, revision, and semantic denotation.

Human scientific observation = PMO + recursive self-model + semantic closure.

The universe therefore contains observation without requiring a universal consciousness, while human consciousness and scientific practice realise a higher-order observer structure. The universe physically registers distinctions; humans can make the machinery of registration itself into a new distinction.

26 Worked System: Stern–Gerlach Measurement

A Stern-Gerlach apparatus provides a concrete physical system in which every stage of PO and PMO can be identified. A beam of neutral atoms with a magnetic moment passes through a nonuniform magnetic field. The spin degree of freedom becomes correlated with spatial motion, and a downstream detector converts the separated wavepackets into stable records. The experiment is routinely used as the canonical model of spin measurement. Modern analyses treat spin and position quantum-mechanically and explicitly study the role of apparatus resolution and environment. [4]

26.0.1 Physical apparatus

- **1. Atomic source:** Produces a beam of atoms and establishes the preparation context.
- **2. Collimator:** Restricts transverse position and momentum so later splitting can be resolved.
- **3. Inhomogeneous magnet:** Selects a spatial axis and couples spin to motion.
- **4. Free-flight region:** Allows the correlated momentum alternatives to separate in position.
- **5. Detection screen:** Transduces atom arrival into local material change.
- **6. Camera/electronics:** Amplifies and stores the spot location as a durable record.
- **7. Human observer:** Models the apparatus, identifies the selected basis, and denotes the result.

26.0.2 Incoming superposition

Choose the magnet's gradient direction as z. An incoming atom may be prepared in

$|\psi\_spin\rangle = \alpha|\uparrow\_z\rangle + \beta|\downarrow\_z\rangle$, $|\alpha|^2+|\beta|^2=1$. (21)

Let $|\phi\_0\rangle$ be the initial spatial wavepacket. The complete incoming state is

$|\Psi\_0\rangle = (\alpha|\uparrow\_z\rangle + \beta|\downarrow\_z\rangle) \otimes |\phi\_0\rangle$. (22)

At this stage, the spin alternatives are admissible and coherently retained. There is not yet a z-record.

26.0.3 Mechanical distinction selected by the magnet

The magnetic interaction is approximately

$H\_int = - \mu\cdot B(r)$. (23)

For a field with z-gradient $B\_z(z) \approx B\_0 + bz$, the two z-spin components experience opposite forces. Over an interaction time τ, the apparatus produces opposite momentum shifts $\pm\Delta p\_z$. The unitary interaction therefore acts as

$|\uparrow\_z\rangle|\phi\_0\rangle \to |\uparrow\_z\rangle|\phi\_+\rangle$, $|\downarrow\_z\rangle|\phi\_0\rangle \to |\downarrow\_z\rangle|\phi\_-\rangle$. (24)

By linearity,

$|\Psi\_1\rangle = \alpha|\uparrow\_z\rangle|\phi\_+\rangle + \beta|\downarrow\_z\rangle|\phi\_-\rangle$. (25)

The magnet has not passively revealed a basis-independent label. Its physical orientation selected the z distinction and converted it into spatially separable channels. Rotating the magnet changes the projectors and therefore changes the mechanical observation performed.

26.0.4 Propagation and spatial resolution

During free flight, the wavepackets separate. A useful discriminability condition is

$|\langle\phi\_-|\phi\_+\rangle| \ll 1$. (26)

When the overlap remains large, the machinery cannot reliably distinguish the alternatives. When it is small, the apparatus has converted spin distinction into a resolvable position distinction. The collimator, field gradient, transit time, atom mass, detector distance, and detector resolution all belong to the observer machinery because all determine whether equation (26) is achieved.

26.0.5 Detection and record formation

Let $|D_0\rangle$ be the ready state of the detector and $|D_+\rangle$, $|D_-\rangle$ be distinct detector records. Detection gives

$\alpha|\uparrow_z\rangle|\phi_+\rangle|D_0\rangle + \beta|\downarrow_z\rangle|\phi_-\rangle|D_0\rangle$ (27a)

$\to \alpha|\uparrow_z\rangle|\phi_+\rangle|D_+\rangle + \beta|\downarrow_z\rangle|\phi_-\rangle|D_-\rangle$. (27b)

The detector then couples to many uncontrolled environmental degrees of freedom: emitted photons, lattice vibrations, electrical currents, memory states, and surrounding matter. Denote the resulting environmental states by $|E_+\rangle$ and $|E_-\rangle$:

$|\Psi_2\rangle = \alpha|\uparrow_z,\phi_+,D_+,E_+\rangle + \beta|\downarrow_z,\phi_-,D_-,E_-\rangle$. (28)

The local coherence between the two detector records is multiplied by

$\gamma = \langle E_-|E_+\rangle$. (29)

For macroscopically distinct records, $|\gamma|$ becomes extremely small. The reduced detector description is then approximately

$\rho_D \approx |\alpha|^2|D_+\rangle\langle D_+| + |\beta|^2|D_-\rangle\langle D_-|$. (30)

26.0.6 Quantum-instrument description

For an ideal z measurement, the effects are

$\Pi_+ = |\uparrow_z\rangle\langle\uparrow_z|$, $\Pi_- = |\downarrow_z\rangle\langle\downarrow_z|$. (31)

$p_+ = \mathrm{Tr}(\rho\Pi_+) = |\alpha|^2$, $p_- = \mathrm{Tr}(\rho\Pi_-) = |\beta|^2$. (32)

The complete real apparatus is more accurately represented by an instrument $\{\mathcal{I}_+,\mathcal{I}_-\}$, because it not only assigns probabilities but also records the conditional state and detector change.

26.0.7 Full semantic breakdown

- **Semantic stage:** Persist **Physical component:** Prepared atomic beam **Mathematical structure:** **What becomes real:** $\Psi_0\rangle$
- **Semantic stage:** Distinguish **Physical component:** Magnet axis and field gradient **Mathematical structure:** $\{\Pi_+,\Pi_-\}$ **What becomes real:** The z-spin alternatives become the enacted question.
- **Semantic stage:** Relate **Physical component:** Magnetic coupling **Mathematical structure:** $H_{int} = -\mu\cdot B$ **What becomes real:** Spin is linked to translational motion.
- **Semantic stage:** Propagate **Physical component:** Magnet plus free flight **Mathematical structure:** U_{SM} and **What becomes real:** $\phi_\pm\rangle$
- **Semantic stage:** Resolve **Physical component:** Wavepacket separation and detector resolution **Mathematical structure:** **What becomes real:** $\langle\phi_-$
- **Semantic stage:** Recurse/Amplify **Physical component:** Detector, environment, electronics **Mathematical structure:** **What becomes real:** $D_\pm E_\pm\rangle$
- **Semantic stage:** Close **Physical component:** Stored spot and conditional state **Mathematical structure:** $\{p_k,r_k,\rho_{S}$ **What becomes real:** k}}
- **Semantic stage:** Reflect/Denote **Physical component:** Human scientific observer **Mathematical structure:** $\Lambda_H(\mathcal{M},k)$ **What becomes real:** The observation machinery itself becomes an explicit model.

Worked-system conclusion The observed spin result is not attributable to the atom alone. It is the closure of the atom's admissible state, the magnet's selected distinction, the spin-position coupling, free propagation, detector resolution, environmental stabilisation, and the retained record. Human observation adds a model of that entire machinery.

Once the conventional mathematics has been displayed, the entire Stern-Gerlach chain can be compressed without losing its denotation:

$Q[\psi] \rightarrow D_n \rightarrow R_{\mu B} \rightarrow G_x \rightarrow V_D \rightarrow K_E \rightarrow C_k \rightarrow \Lambda_H$. (33)

The symbols denote

- **Q[ψ] — Coherent unresolved admissibility:** Incoming spin state
- **D_n — Mechanical selection of distinction n:** Magnet orientation
- **R_{μB} — Physical relation that couples alternatives to channels:** Magnetic interaction
- **G_x — Propagation into an observable carrier:** Spatial separation
- **V_D — Resolution by observer machinery:** Detector discriminability
- **K_E — Recursive retention and environmental stabilisation:** Pointer record plus decoherence
- **C_k — Closure into outcome k and conditional state:** Stored upper or lower detection
- **Λ_H — Reflexive semantic denotation:** Human report and mathematical model

This compression is valid only because every class has been denoted once against the expanded mechanics. It is therefore not a replacement for the mathematics but a reusable relational language for its physical necessity and order.

27 Reality as Retained Distinction

27.0.1 Superposition is related to reality through observation machinery

Superposition is not a pre-reality fiction. It is the unresolved physical account retained before a mechanically stable distinction exists for a given observer. Mechanical observation maps that account into a family of possible records and conditional states:

$\rho_S \xrightarrow{\mathcal{M}_O} \{p_k, r_k, \rho_{S|k}\}$. (34)

27.0.2 Reality is observer-relative without being arbitrary

A record is relative to an observer architecture, but it is not freely invented. The state, coupling, conservation laws, detector thresholds, and environmental dynamics constrain what records can occur. Relativity here means physically indexed, not subjective whim.

27.0.3 Mechanical observation does not require consciousness

The detector record can form before any human reads it. Human consciousness enters as a higher-order, reflexive, semantically expressive observer architecture. This preserves a physical account of measurement while giving human observation genuinely additional structure.

27.0.4 Decoherence is not silently equated with a complete interpretation

The paper uses decoherence to explain stable pointer records and effective loss of locally accessible interference. It does not assume that decoherence alone settles every question about unique global outcomes. The Principles of Observation and Mechanical Observation are intended as common operational machinery beneath competing interpretations, not as an unsupported declaration that all interpretive questions have vanished.

27.0.5 Testable use of the framework

The principles become scientifically productive when they expose a missing observer component, predict how changing observer architecture changes the measurable distinction, or prove that two apparently equivalent measurement descriptions do not preserve the same complete logical action. Candidate tests include imperfect detector resolution, weak measurement, delayed readout, measurement-chain reversibility, and quantum-reference-frame changes.

PO - universal observation A distinction becomes physically accountable relative to another system through relation, persistence, and closure.

$\mathfrak{O}_{O\leftarrow S} = \mathcal{C}_O \circ \mathcal{P}_O \circ \mathcal{R}_{OS} \circ \mathcal{D}_O(S)$.

PMO - structured observation An observer architecture converts admissible system distinctions into stable records through prepared machinery, channel-defining coupling, propagation, resolution, amplification, and record production.

$\mathfrak{O}^{mech}_{O\leftarrow S} = \mathcal{C}_O \circ \mathcal{R}ec_O \circ \mathcal{A}_O \circ \mathcal{P}roj_O \circ \mathcal{U}_{SM}$.

Human reflexive observation A mechanical observation whose record and machinery can be recursively modelled and semantically denoted by the observer.

$\mathfrak{O}_H^{(2)}(S) = \Lambda_H \circ \mathrm{Rec}_H \circ \mathfrak{O}_{H\leftarrow S}^{mech}$.

The complete proposed quantum-to-reality operation is

Admissible logical actions → coherent state → observer-selected distinction → system-machinery entanglement → propagated record channels → decoherent stabilisation → observer-relative fact → reflexive semantic model. (35)

The next mathematical work is to:

1. Classify observer architectures by the distinctions they can physically enact.

2. Define equivalence of observation machinery under the Principle of Equivalent Action.
3. Use the Principle of Logical Action to distinguish apparatuses that give the same probabilities but different disturbance, memory, topology, or causal order.
4. Develop a relational normal form for quantum instruments and measurement chains.
5. Apply the framework to continuous measurement, field detection, gravitational observation, and human sensory systems.
6. Test whether the observer hierarchy provides a common bridge between quantum measurement, relational state attribution, and classical record formation.

Final statement Superposition is unresolved admissible reality. Observation is the production of a retained distinction relative to another system. Mechanical observation is the physical machinery that makes the distinction discriminable and persistent. Human scientific observation is the recursive capacity to make that machinery itself observable and mathematically denotable.

27.1 The complete quantum-to-reality chain

The observer machinery completes the action sequence:

admissibility	→ superposition → mechanical distinction → system-apparatus correlation → propagation through machinery → record stabilisation → observer-relative fact → semantic closure.

Superposition is unresolved physical reality. Mechanical observation does not transform unreality into reality. It transforms an unresolved relational account into a persistent distinction available to another system. Human observation then re-enters the operation, models the machinery, and denotes the result as language, diagram, or equation.

Part VII — The Unified Relational Compiler

28 The Framework as a Physical Grammar of Mathematics

The preceding parts establish one repeated fact: established equations can be unfolded into the same operational dependency structure. The framework's strength is therefore not merely a new collection of equations. It is the ability to compile ordinary mathematics into a relational language that preserves why the mathematics must possess its form.

Define a **relational compiler**

$$\mathfrak{C}:\text{conventional expression} \to (\text{semantic trace},\text{equivalence class},\text{closure form}).$$

For an expression E, the compiler performs the following operations.

1. **Locate the retained whole.** What system must remain one account?
2. **Identify distinctions.** What alternatives, subsystems, frames, states, or channels are non-interchangeable?
3. **Identify relation.** What operator, coupling, connection, or constraint makes those distinctions mutually reachable?
4. **Recover order.** Which operations must precede which others, and which commutations are physically valid?
5. **Construct the action class.** Which written descriptions share the same invariant dynamics under PEA?
6. **Retain logical subclasses.** Which histories differ in phase, topology, boundaries, measure, or observer architecture under PLA?
7. **Specify propagation.** How is the relation enacted through time, space, scale, or composition?
8. **Specify resolution.** What cancels, decoheres, quotients, projects, or becomes observationally distinct?
9. **Specify return.** What feedback, recurrence, memory, or environmental record keeps consequences inside the system?
10. **Close and compress.** What equation is the invariant normal form of the complete traversal?

A valid compressed class is reusable only after its conventional derivation has been shown once. Thereafter the class can denote the same physical necessity in later calculations without repeating every intermediate line.

29 A Unified Class Ledger

The following class system integrates the recent papers.

- $S0$ — **Systemhood:** $P \to D \to R \to G \to V \to K \to C$
- $S1$ — **Recoverable object:** $O = [\tau]_{\sim}$
- $A0$ — **Equivalent Action:** $S_1 \sim_{EA} S_2 \Leftrightarrow \mathcal{E}(S_1) = \mathcal{E}(S_2)$
- $A1$ — **Logical Action:** $S_1 \sim_{LA} S_2 \Leftrightarrow \Lambda(S_1) \cong_{ord} \Lambda(S_2)$
- $A2$ — **Complete physical equivalence:** EA + LA + quantum + observational closure
- $F0$ — **Local field:** $x \mapsto \Phi(x)$
- $F1$ — **Relational transport:** $D_\mu = \partial_\mu + A_\mu$
- $F2$ — **Curvature / remainder:** $F_{\mu\nu} = [D_\mu, D_\nu]$
- $Q0$ — **Admissibility:** $\mathcal{A}(I)$
- $Q1$ — **Equivalent-action quotient:** $Q(I) = \mathcal{A}(I)/\sim_{EA}$
- $Q2$ — **Coherent retention:** $
- $Q3$ — **Action phase:** $w[\alpha] = e^{iS[\alpha]/\hbar}$
- $Q4$ — **Relational overlap:** $
- $Q5$ — **Closed probability:** $p(E
- $Q6$ — **Closed propagation:** $U^\dagger U = I, U(t) = e^{-iHt/\hbar}$
- $Q7$ — **Composition:** $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$
- $O0$ — **Relational observation:** retained correlation $\mathfrak{O}_{O \leftarrow S}$
- $O1$ — **Mechanical observation:** quantum instrument $\{\mathcal{I}_k\}$
- $O2$ — **Record:** persistent discriminable state r_k
- $O3$ — **Reflexive observation:** observer models its observing machinery
- $O4$ — **Semantic denotation:** record → language, model, equation
- $W0$ — **Worldsheet action:** $S_{\mathrm{P}}[X, h]$
- $W1$ — **Topological action class:** sum over Σ and gauge quotient
- $G0$ — **Geometric closure:** $G_{\mu\nu} = 8\pi G T_{\mu\nu}^{\mathrm{tot}}$

This ledger is not intended to replace standard notation. It supplies a faster relational index. Once a class has been fully defined, a derivation can reference its operation rather than restating its philosophy.

30 Formal Proposition Ledger

30.1 Proposition 1 — No object before recoverability

A symbol can be introduced provisionally, but physical objecthood is justified only when a distinction remains recoverable through an admissible relation and closure.

$$O = [\tau]_{\sim}.$$

30.2 Proposition 2 — Denotational persistence

If a semantic type maps compositional operations across multiple domains and cannot be removed without making later operations undefined, it is not merely an optional interpretation of those domains. It is a candidate invariant grammar of their construction.

30.3 Proposition 3 — Equivalent Action fibre theorem

For the dynamics map $\mathcal{E}: \mathcal{A} \to \mathcal{V}$, each fibre $\mathcal{E}^{-1}(L)$ is an Equivalent Action class. Distinct written actions in one fibre are representatives of the same invariant governing dynamics.

30.4 Proposition 4 — Logical refinement

If two action representatives have equal dynamics but non-isomorphic ordered traces, then they are PEA-equivalent but not PLA-equivalent. PLA therefore refines, rather than contradicts, PEA.

30.5 Proposition 5 — Unresolved action retention

If two logical actions are admissible and no physical resolving distinction has occurred, deletion of either action introduces information not supplied by the system. The complete unresolved account must retain both.

30.6 Proposition 6 — Superposition normal form

When unresolved actions must be retained jointly, composed distributively, and allowed to interfere before resolution, their minimal conventional normal form is a linear combination over action classes.

30.7 Proposition 7 — Interference as retained relation

The cross term

$$2\mathrm{Re}(c_1 c_2^*)$$

is the mathematical residue of a relation between unresolved logical actions. A classical mixture is obtained when this relation is lost or ignored.

30.8 Proposition 8 — Mechanical observation transfer

A unitary measurement coupling does not initially delete superposition. It transfers unresolved alternatives into correlations between system and observer machinery:

$$\left(\sum_i c_i \, |s_i\rangle\right) \Big| M_0\rangle \to \sum_i c_i \Big| s_i\rangle |M_i\rangle.$$

30.9 Proposition 9 — Record persistence

A mechanical state M_k is a record when its distinction from alternative records remains recoverable under the relevant future dynamics.

30.10 Proposition 10 — Gauge curvature as logical remainder

Curvature is the residual difference produced when relational transport through two ordered local paths fails to close identically.

30.11 Proposition 11 — String-geometric closure

If a quantum worldsheet is to remain Weyl-consistent, its background spacetime fields must satisfy the vanishing-beta-function conditions. Gravitational field equations are therefore the spacetime closure of the extended quantum action.

30.12 Proposition 12 — Universal system scope

Because the universe is a system and the semantic grammar denotes systemhood, the framework applies universally in scope. Domain-specific derivation concerns resolution, not reach.

31 What Monograph II Establishes

The complete chain is now visible:

fixed self-description	→ distinction–identity–relation → semantic systemhood → equivalent action classes
	→ ordered logical action → field transport and curvature → bounded possibility → quantum admissibility
	→ superposition and interference → mechanical observation → extended string action → gravitational cl
	→ human reflexive denotation → mathematical normal form.

The unity is not produced by declaring all entities identical. The framework preserves the differences among particle, field, apparatus, observer, string, and geometry while showing that each is a stage or realisation of one system grammar.

The deepest result of the monograph is therefore:

> ***Reality executes one continuous logical action at multiple resolutions. Quantum mechanics retains its admissible possibilities, string theory extends its traversals, general relativity closes its shared geometry, mechanical observation retains its facts, and mathematics compresses the completed invariants.***

32 Research Programme

The next stage is not to broaden the framework beyond systemhood. It is to increase derivational resolution.

32.1 Action programme

1. Classify null Lagrangians and boundary-active representatives under PEA and PLA.
2. Formalise the inverse variational map from governing operators to action classes.
3. Develop categorical or rewriting representations of ordered logical traces.
4. Integrate ATOI notation with path ordering and causal composition.

32.2 Quantum programme

1. State the coherent-retention axioms in minimal independent form.
2. Prove the scalar-field choice under explicit continuity, composition, and tomography assumptions.
3. Complete the Born-rule derivation for all finite dimensions using generalised effects.
4. Classify density operators as observational quotients of logical-action ensembles.
5. Determine when environmental dispersal changes a superposition class into a mixture class.

32.3 Field and gravity programme

1. Derive connection structure directly from preserved local comparability.
2. Classify curvature as a failure of ordered closure in Abelian and non-Abelian settings.
3. Derive a universal total-field stress-energy closure and compare its low-energy form with Einstein gravity.
4. Connect Monograph I's partition channels to field-sector decompositions without importing labels prematurely.

32.4 String programme

1. Construct the PEA class containing Nambu–Goto and Polyakov representatives.
2. Compute the PLA trace of gauge fixing, ghost measure, topology, and anomaly cancellation.
3. Interpret left-right mover closure through the whole-part-boundary ledger.
4. Derive the low-energy spacetime action as the closure normal form of worldsheet consistency.

32.5 Observation programme

1. Classify observer architectures by the distinctions they can physically retain.
2. Define equivalence and inequivalence of quantum instruments under PEA and PLA.
3. Distinguish record formation, decoherence, outcome conditioning, and human semantic denotation.
4. Extend the Stern–Gerlach analysis to cavities, interferometers, and field detectors.

32.6 Compression programme

1. Build a verified dictionary from conventional derivations to relational classes.
2. Require every compression to preserve expansion, composition, and failure conditions.
3. Develop a formal compiler capable of detecting missing distinction, relation, order, resolution, or closure terms in an equation.
4. Use the compiler to compare theories at the level of logical action rather than notation alone.

33 Conclusion

Monograph I established a world of distinctions. Monograph II establishes what those distinctions must do.

They must remain recoverable. They must relate without becoming identical. Their relations must propagate. Unresolved alternatives must remain jointly available until a physical distinction resolves them. The products of action must return into a continuing system if they are to become meaningful. Observers must possess machinery capable of converting distinctions into records. Mathematical equations must close the entire account without erasing distinctions that can still affect later action.

From these requirements emerge the central structures of modern physics: action equivalence, path order, gauge connection, curvature, quantum superposition, complex phase, Hilbert space, unitary evolution, measurement instruments, string worldsheets, and gravitational geometry.

The framework therefore does not sit beside reality as commentary. It maps the logical action by which reality becomes physically and mathematically resolvable.

Reality executes the operation.	Observation retains the distinction.	Mathematics records the closure.

Appendix A — Canonical Notation

- φ**:** self-description fixed point satisfying $\varphi^2 = \varphi + 1$
- r**:** retained remainder parameter $1/(2\varphi)$
- ν**:** complementary amplitude $1 - r$
- $\mathfrak{G}$**:** seven-stage semantic grammar (P, D, R, G, V, K, C)
- $\mathbf{R}, \wp, \mathbf{O}$**:** compact Rationality, Realisation, Observation operators
- τ**:** ordered operational traversal
- $[\tau]_{\sim}$**:** equivalence class defining recoverable objecthood
- $\mathcal{E}$**:** Euler-Lagrange or governing-dynamics map
- $\sim_{EA}$**:** Equivalent Action relation
- $\Lambda(S)$**:** ordered Logical Action trace
- $\sim_{LA}$**:** Logical Action equivalence
- D_μ**:** covariant derivative / relational transport
- $F_{\mu\nu}$**:** curvature / nonclosure remainder
- $\mathcal{A}(I)$**:** admissible logical actions from initial condition I
- $\mathcal{Q}(I)$**:** quotient of admissible actions by PEA
- $\mathfrak{O}$**:** observation map
- $\mathcal{M}_O$**:** observer architecture
- $\mathcal{I}_k$**:** quantum instrument operation for record k
- Λ_H**:** human reflexive semantic denotation

Appendix B — Standard Equations under the Semantic Grammar

- $\varphi^2 = \varphi + 1$**:** self-similar relation closes under its own partition
- $v^2 + 2vr + r^2 = 1$**:** direct, relational, and retained histories exhaust the whole
- $S_1 \sim_{EA} S_2$**:** different inscriptions preserve one invariant dynamics
- $D_\mu = \partial_\mu + A_\mu$**:** local distinction requires transport machinery for comparison
- $F_{\mu\nu} = [D_\mu, D_\nu]$**:** ordered transport fails to close identically
- **$:** =_i c_i
- $e^{iS/\hbar}$**:** an action history retains orientation through complex phase
- **$i_t:** =H
- **$p(E{:}{*}{*}\, \rho) = \mathrm{Tr}(\rho E)$
- $\rho \to \mathcal{J}_k(\rho)$**:** observer machinery converts unresolved action into a conditioned record
- $S_{\mathrm{NG}} \sim_{EA} S_{\mathrm{P}}$**:** area and auxiliary-metric descriptions preserve one classical string dynamics
- $\beta^G_{\mu\nu} = 0$**:** worldsheet quantum action closes only on compatible spacetime geometry
- $G_{\mu\nu} = 8\pi G T_{\mu\nu}$**:** shared geometry closes against total field content

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